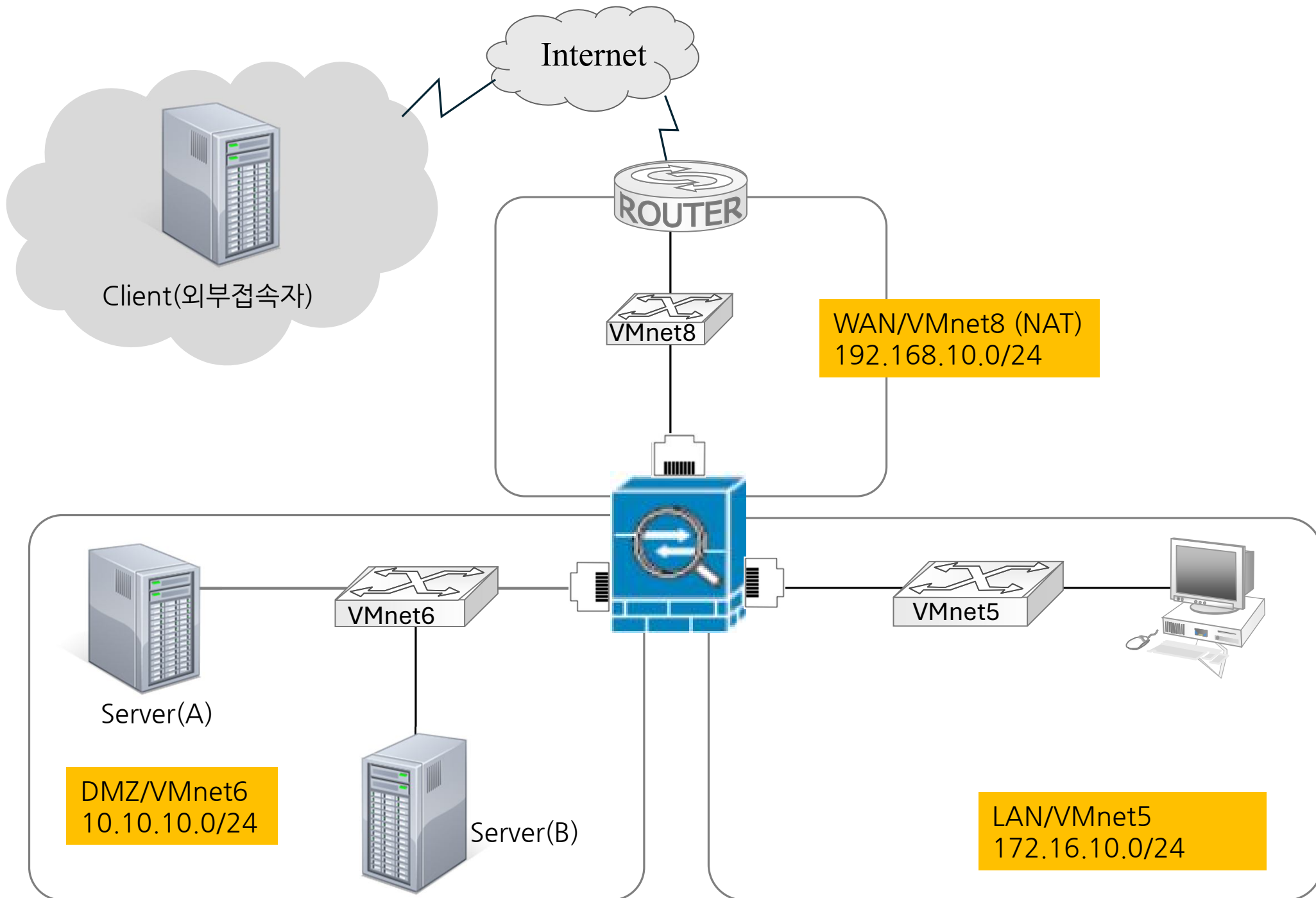


pfSense UTM 설치& 운영



Edit > Virtual Network Editor

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet5	Custom	-	-	-	172.16.10.0
VMnet6	Custom	-	-	-	10.10.10.0
VMnet8	NAT	NAT	Connected	-	192.168.10.0

Add Network... Remove Network Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Settings...

☐ NAT (shared host's IP address with VMs)
NAT Settings...

☒ Host-only (connect VMs internally in a private network)

☐ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet5

☐ Use local DHCP service to distribute IP address to VMs
DHCP Settings...

Subnet IP: Subnet mask:

Restore Defaults Import... Export... OK Cancel Apply Help

Virtual Network Editor

Name	Type	External Connection	Host Connection	DHCP	Subnet Address
VMnet5	Custom	-	-	-	172.16.10.0
VMnet6	Custom	-	-	-	10.10.10.0
VMnet8	NAT	NAT	Connected	-	192.168.10.0

Add Network... Remove Network Rename Network...

VMnet Information

☐ Bridged (connect VMs directly to the external network)
Bridged to: Automatic Settings...

☐ NAT (shared host's IP address with VMs)
NAT Settings...

☒ Host-only (connect VMs internally in a private network)

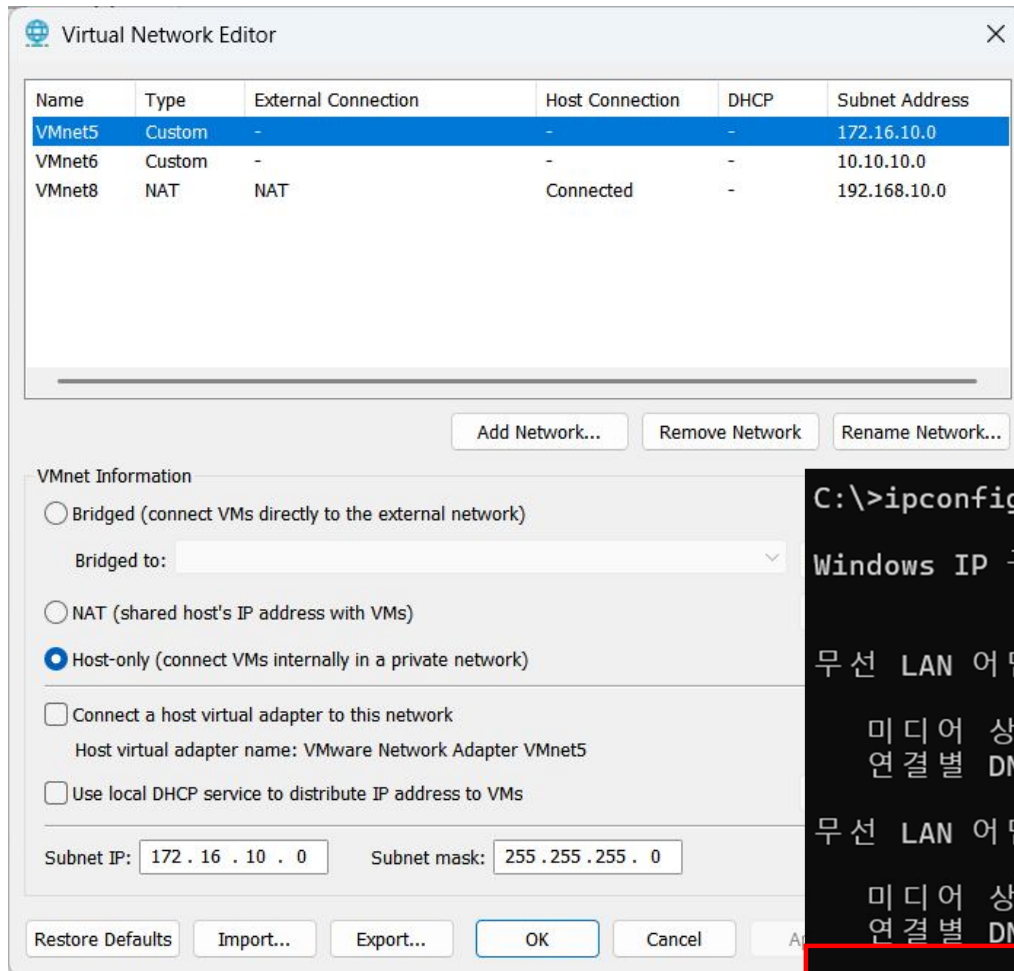
☐ Connect a host virtual adapter to this network
Host virtual adapter name: VMware Network Adapter VMnet6

☐ Use local DHCP service to distribute IP address to VMs
DHCP Settings...

Subnet IP: Subnet mask:

Restore Defaults Import... Export... OK Cancel Apply Help

Edit > Virtual Network Editor



```
C:\>ipconfig
```

Windows IP 구성

무선 LAN 어댑터 로컬 영역 연결* 1:

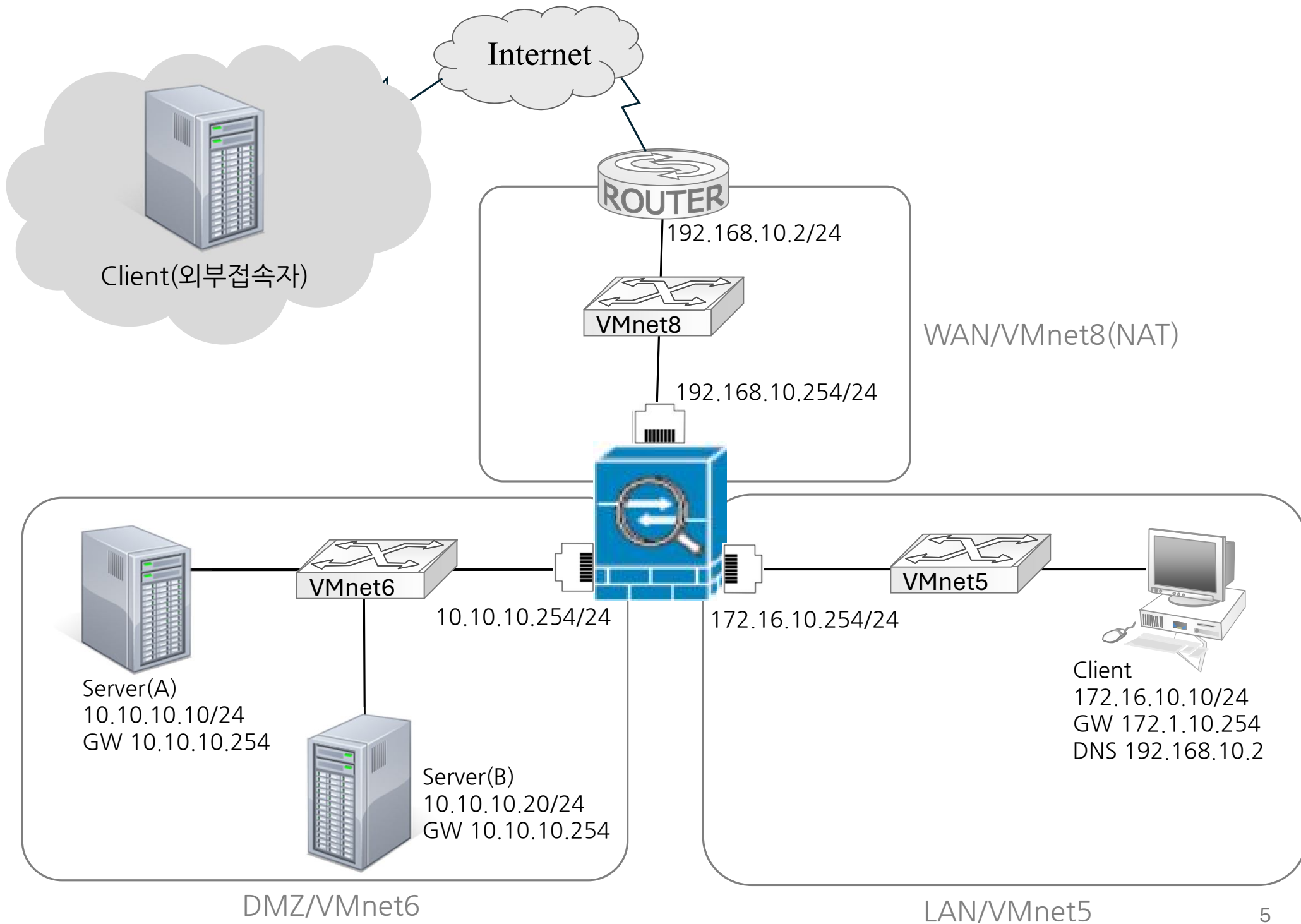
미디어 상태 : 미디어 연결 끊김
연결별 DNS 접미사 :

무선 LAN 어댑터 로컬 영역 연결* 10:

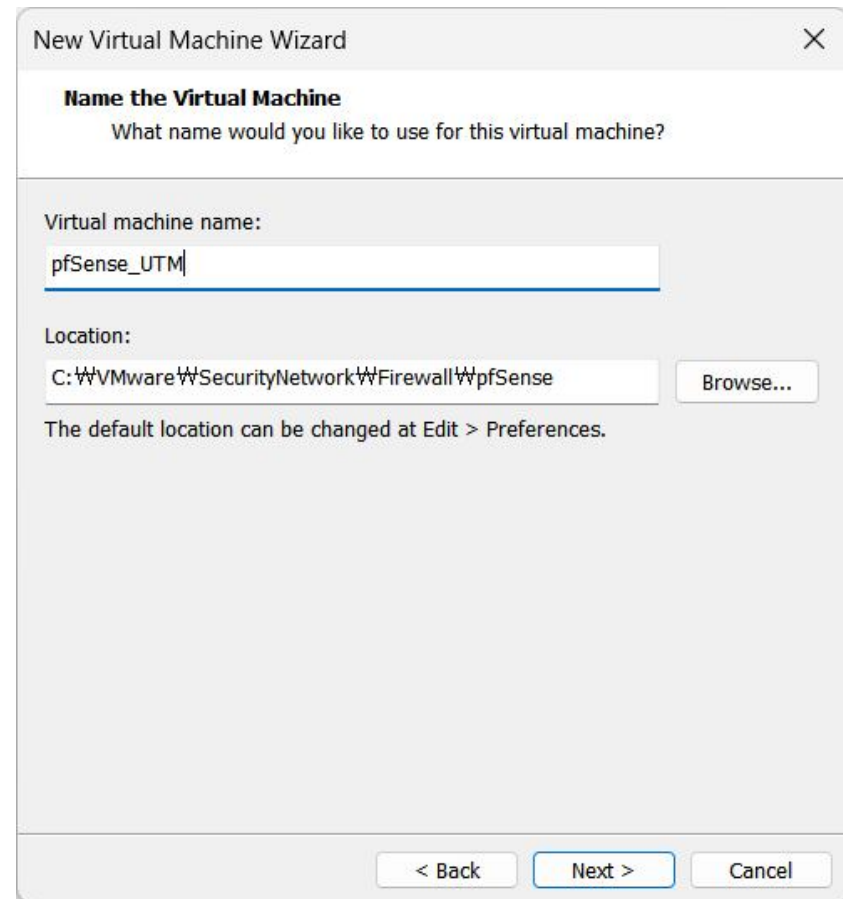
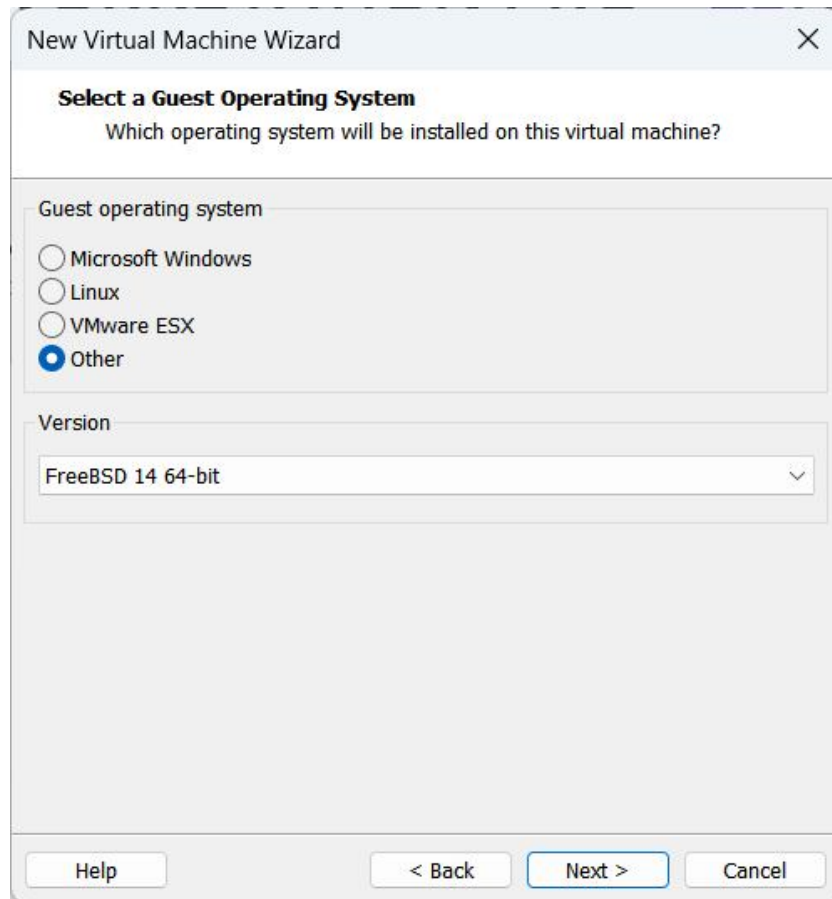
미디어 상태 : 미디어 연결 끊김
연결별 DNS 접미사 :

이더넷 어댑터 VMware Network Adapter VMnet8:

연결별 DNS 접미사 :
링크-로컬 IPv6 주소 : fe80::52ff:3b7:f5d7:484%15
IPv4 주소 : 192.168.10.1
서브넷 마스크 : 255.255.255.0
기본 게이트웨이 :

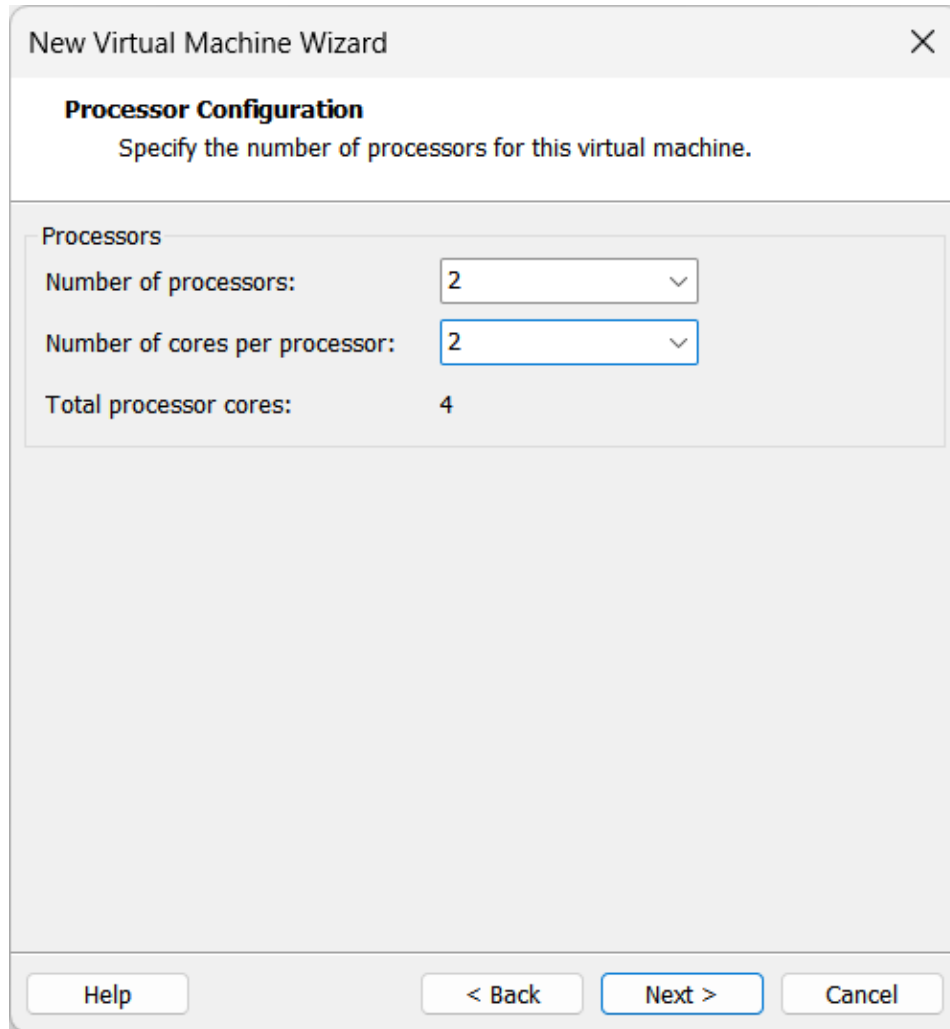


1) Virtual machine 생성



① Version FreeBSD 14 64-bit

Virtual machine 생성



New Virtual Machine Wizard

Processor Configuration
Specify the number of processors for this virtual machine.

Processors

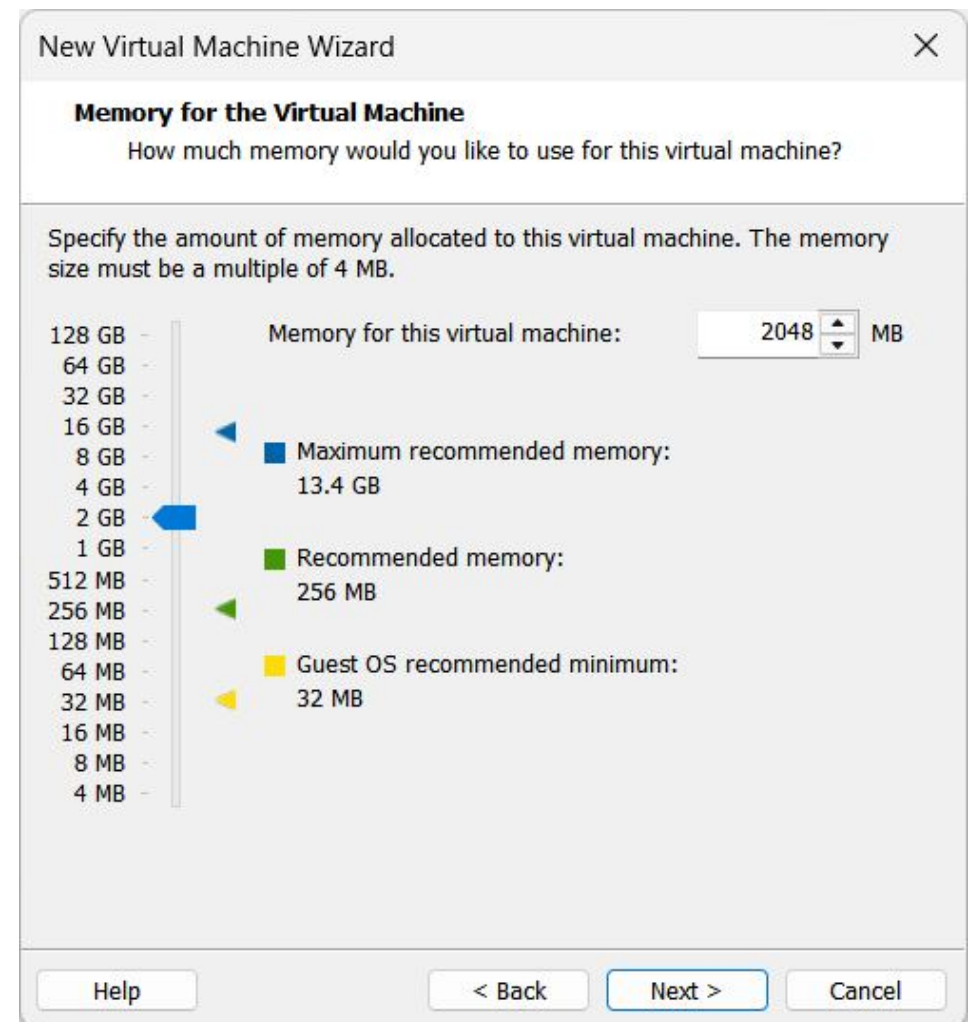
Number of processors: 2

Number of cores per processor: 2

Total processor cores: 4

Help < Back Next > Cancel

② Processor cores : 4



New Virtual Machine Wizard

Memory for the Virtual Machine
How much memory would you like to use for this virtual machine?

Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.

Memory for this virtual machine: 2048 MB

128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB

Maximum recommended memory: 13.4 GB

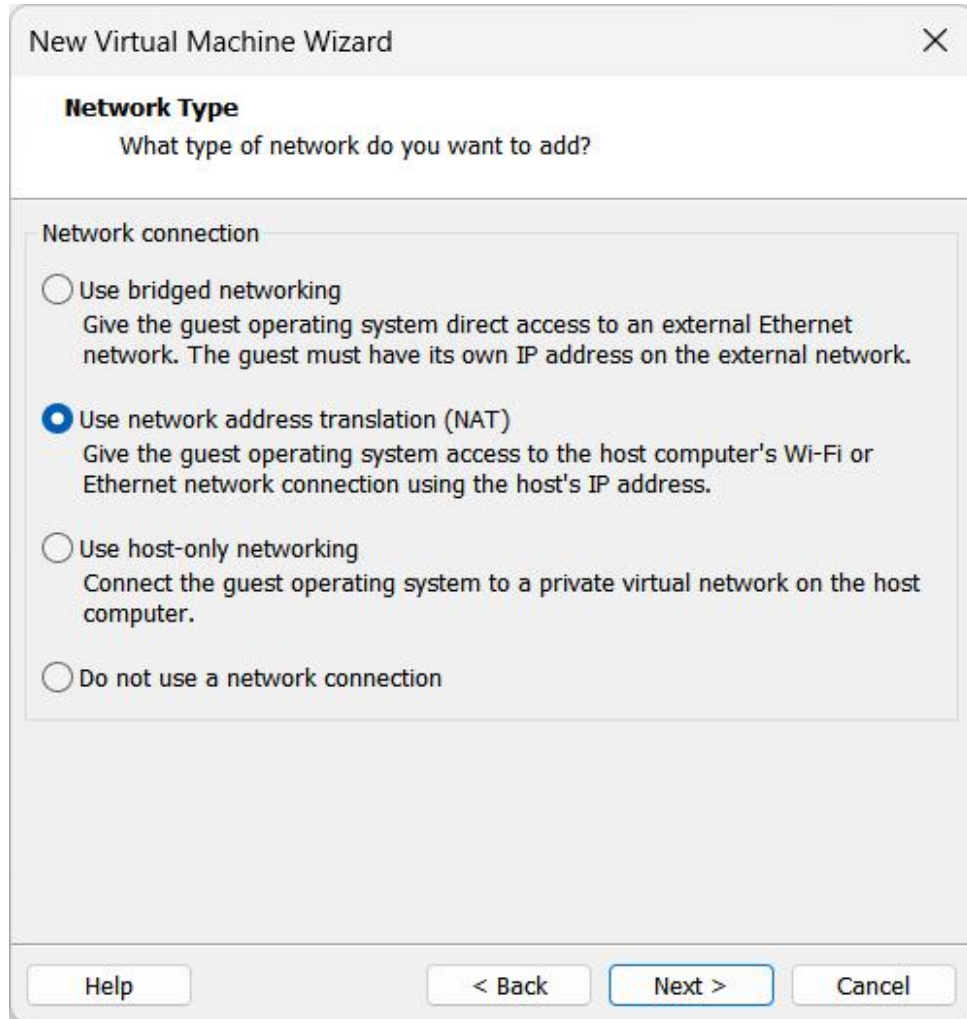
Recommended memory: 256 MB

Guest OS recommended minimum: 32 MB

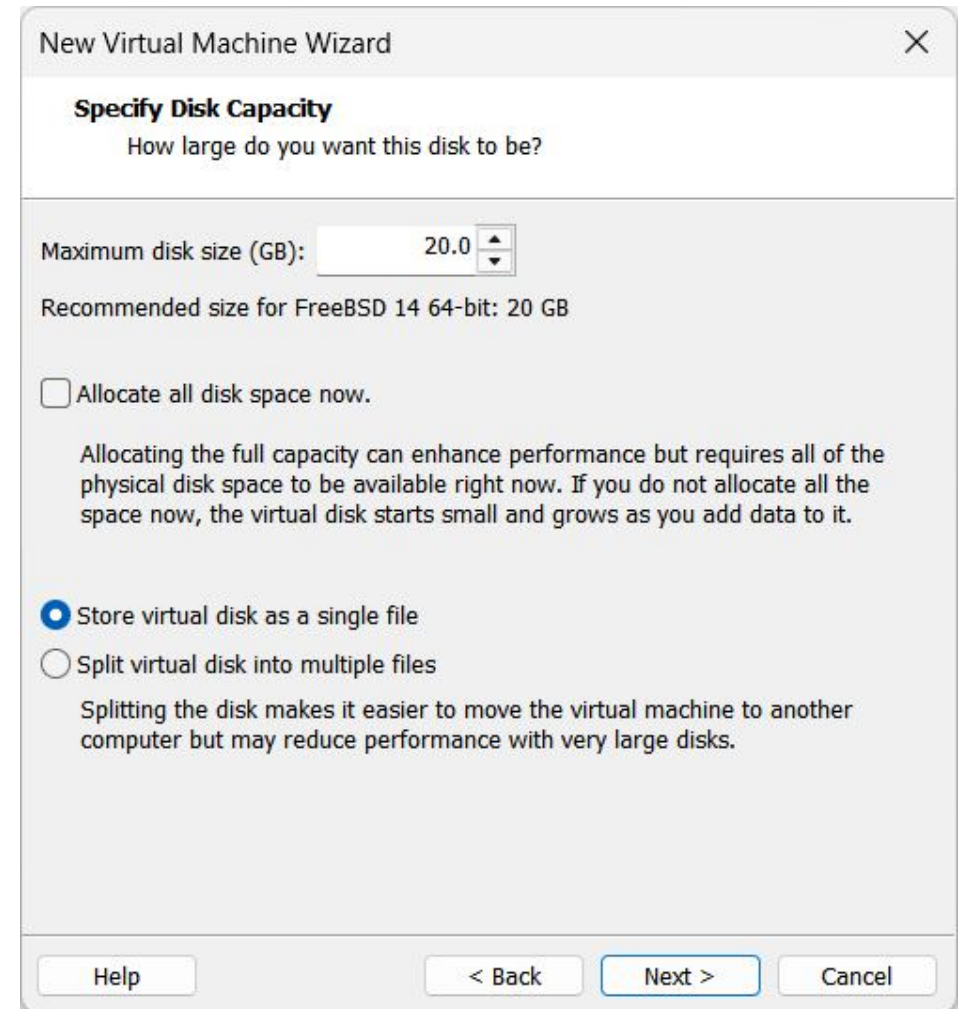
Help < Back Next > Cancel

③ RAM : 2GB

Virtual machine 생성

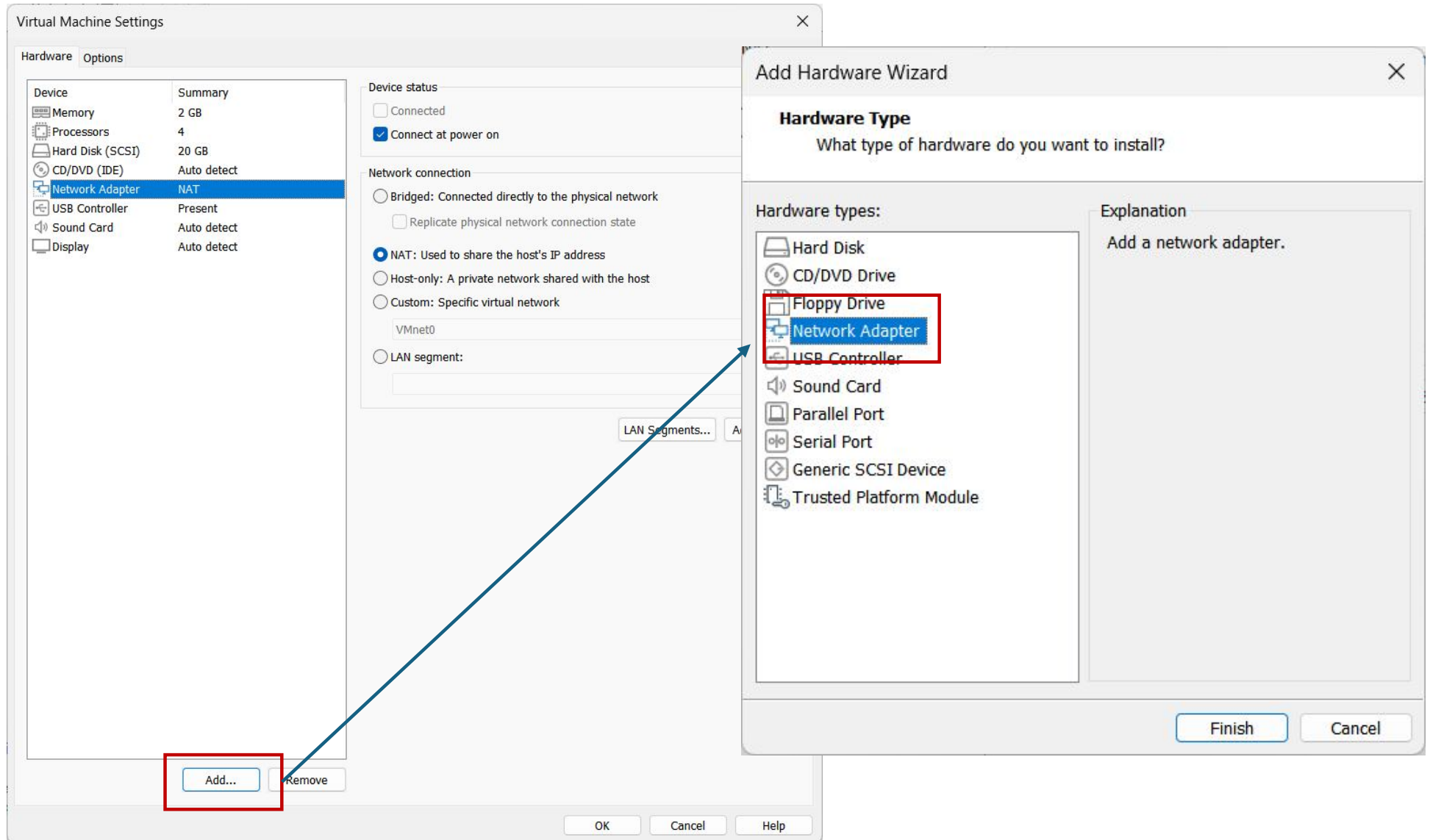


④ Network Type : NAT

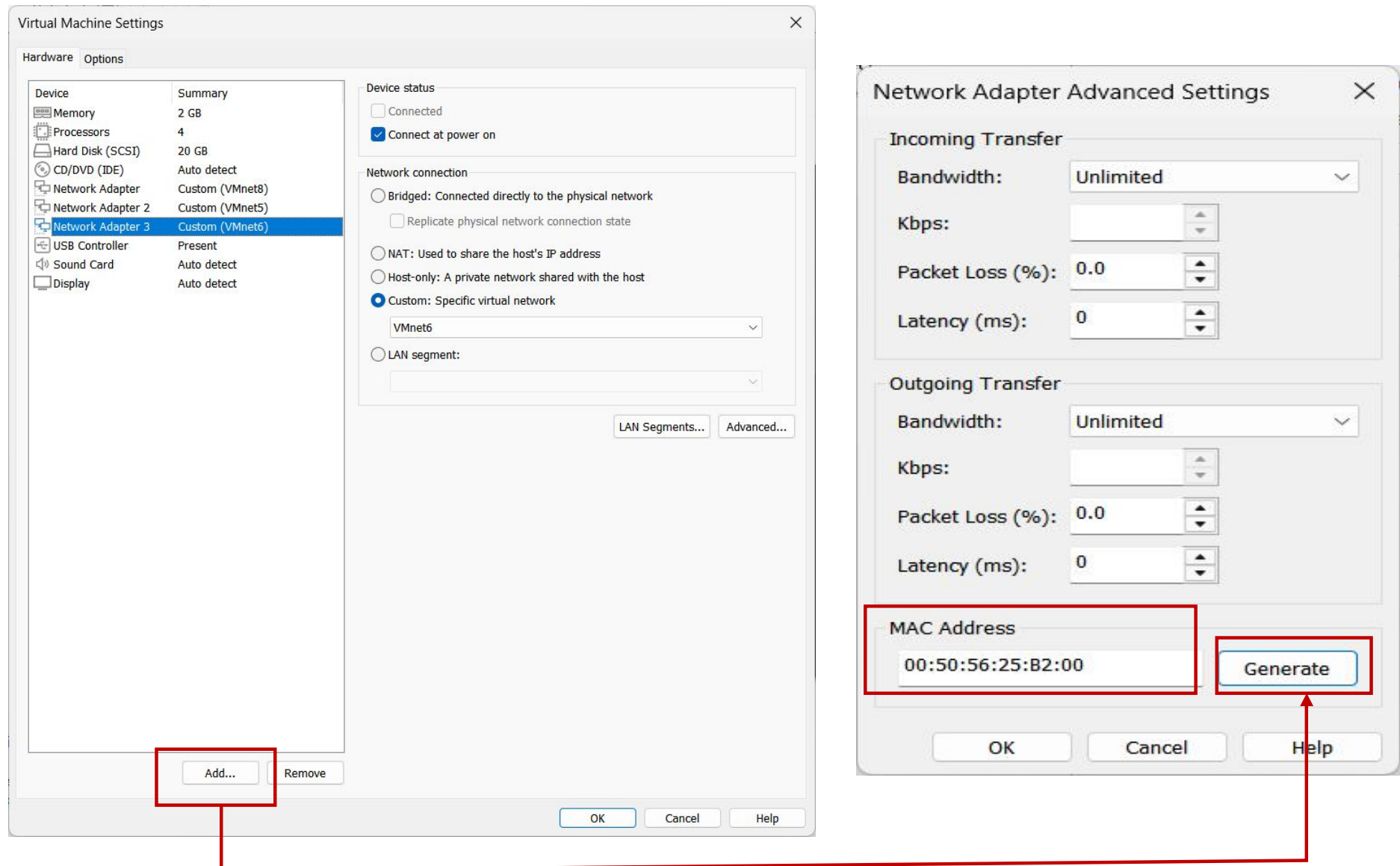


⑤ HardDisk : 20GB

LAN Card 추가 후 MAC 주소 생성과 확인



LAN Card 추가 후 MAC 주소 생성과 확인

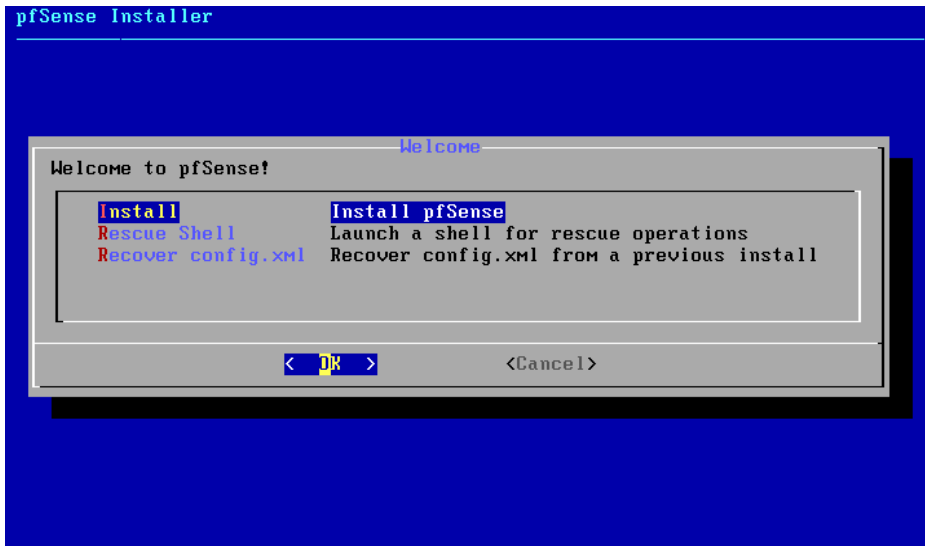


⑥LAN card 추가 후 생성된 LAN card의 MAC 주소 기록

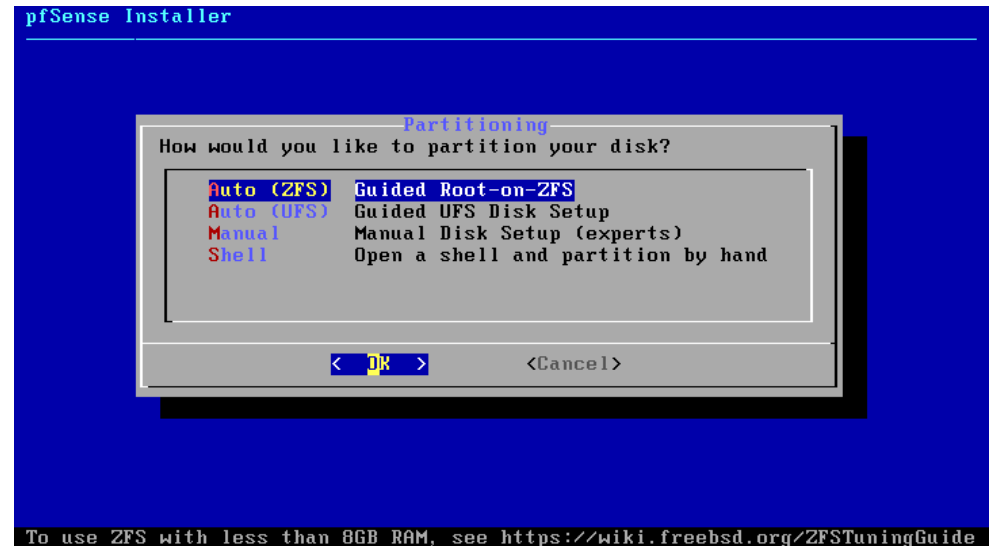
2) pfSense IOS 설치



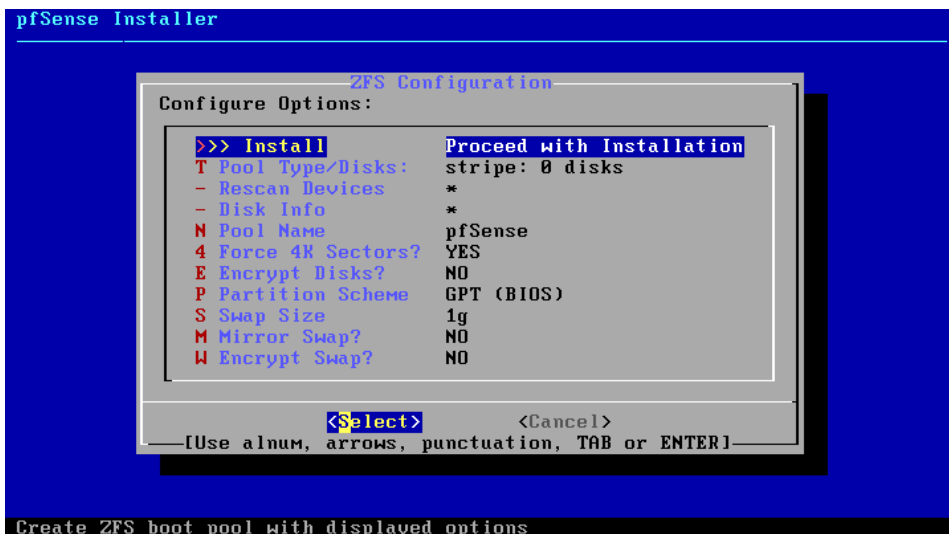
① Accept



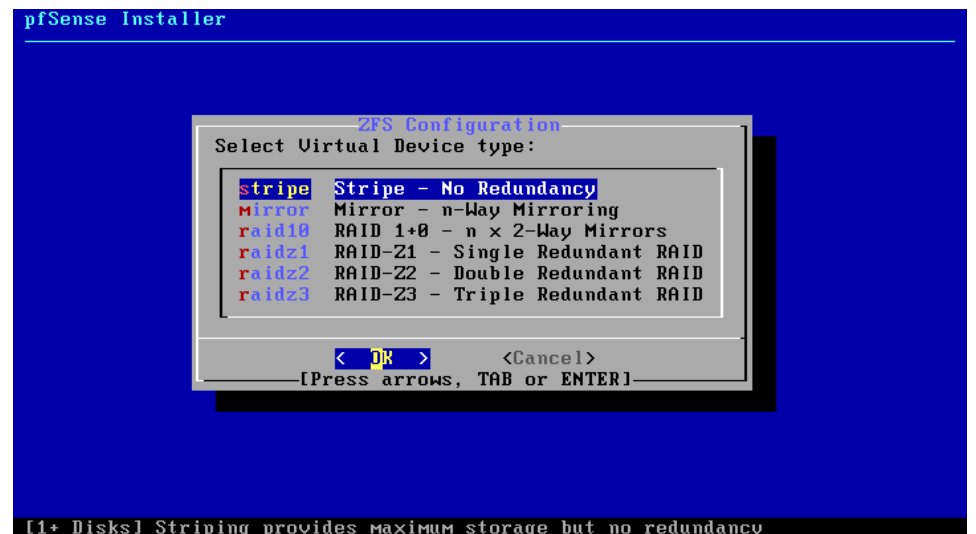
② Install pfsense



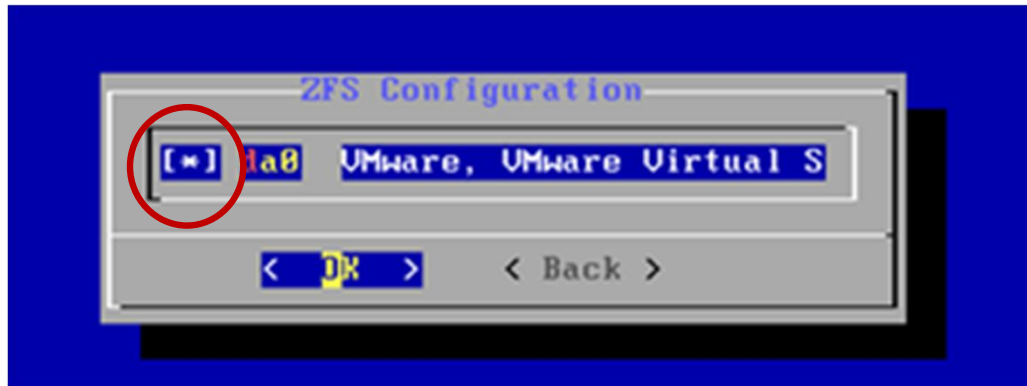
③ Auto(ZFS)



④ Install



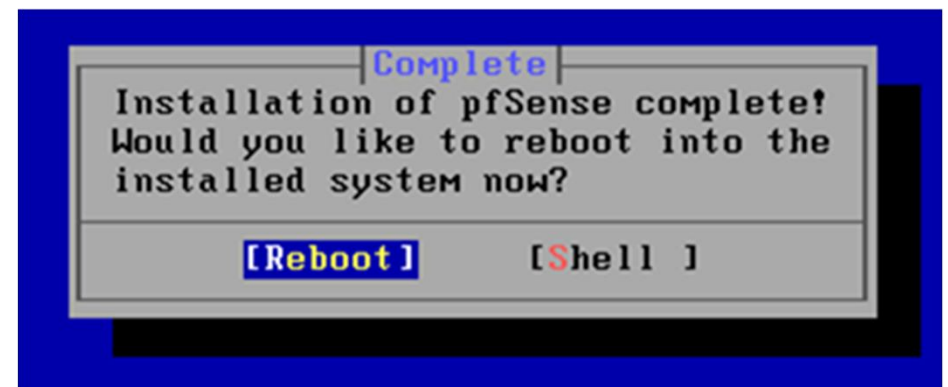
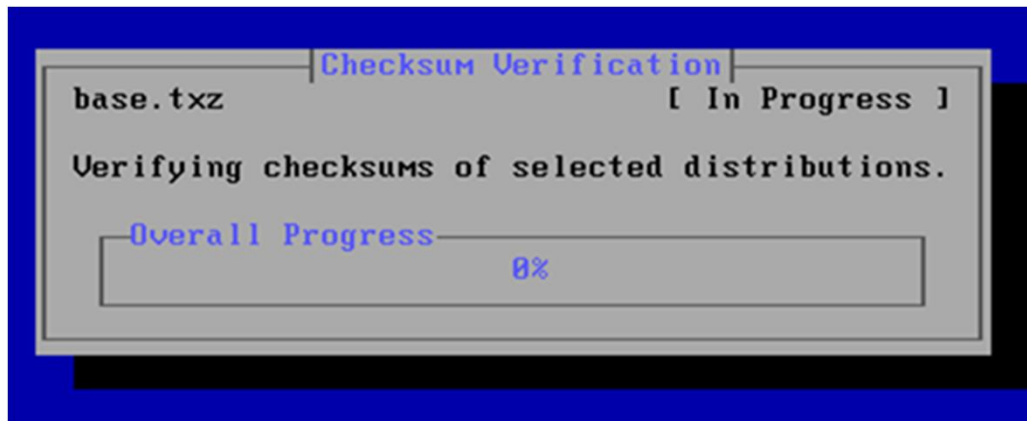
⑤ stripe



⑥ * 체크 후 OK



⑦ Yes



⑧ Reboot

3) pfSense 네트워크 인터페이스 확인

```
done.  
Starting CRON... done.  
pfSense 2.7.2-RELEASE amd64 20231206-2010  
Bootup complete  
  
FreeBSD/amd64 (pfSense.home.arp) (ttyv0)  
  
VMware Virtual Machine - Netgate Device ID: abfef3ef2b25954e3af8  
  
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***  
  
WAN (wan)      -> em0      ->  
LAN (lan)      -> em1      -> v4: 192.168.1.1/24  
  
0) Logout (SSH only)          9) pfTop  
1) Assign Interfaces          10) Filter Logs  
2) Set interface(s) IP address 11) Restart webConfigurator  
3) Reset webConfigurator password 12) PHP shell + pfSense tools  
4) Reset to factory defaults    13) Update from console  
5) Reboot system              14) Enable Secure Shell (sshd)  
6) Halt system                15) Restore recent configuration  
7) Ping host                  16) Restart PHP-FPM  
8) Shell  
  
Enter an option: 1
```

```
Valid interfaces are:  
  
em0      08:50:56:3a:9f:de   (up) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)  
em1      08:50:56:24:1a:79   (up) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)  
em2      08:50:56:25:b2:00 (down) Intel(R) Legacy PRO/1000 MT 82545EM (Copper)  
  
Do VLANs need to be set up first?  
If VLANs will not be used, or only for optional interfaces, it is typical to  
say no here and use the webConfigurator to configure VLANs later, if required.  
Should VLANs be set up now [y:n]? █
```

기록해 놓은 랜카드 MAC주소와 인터페이스 이름 확인

4) pfSense 칩셋 변경 (e1000 → vmxnet3)

```
pfSense_UTM.vmx
파일 편집 보기
scsi0:0.fileName = "pfSense_UTM.vmdk"
scsi0:0.present = "TRUE"
ide1:0.deviceType = "cdrom-image"
ide1:0.fileName = "C:\VMware\SecurityNetwork\설치자료\WpfSense-CE-2.7.2-RELEASE-amd64.iso"
ide1:0.present = "TRUE"
usb.present = "TRUE"
ehci.present = "TRUE"
ethernet0.connectionType = "custom"
ethernet0.addressType = "static"
ethernet0.virtualDev = "e1000"
ethernet0.present = "TRUE"
extendedConfigFile = "pfSense_UTM.vmx"
floppy0.present = "FALSE"
ethernet0.vnet = "VMnet8"
ethernet0.displayName = "VMnet8"
ethernet2.connectionType = "custom"
ethernet2.addressType = "static"
ethernet2.vnet = "VMnet6"
ethernet2.displayName = "VMnet6"
ethernet2.virtualDev = "e1000"
ethernet1.connectionType = "custom"
ethernet1.addressType = "static"
ethernet1.vnet = "VMnet5"
ethernet1.displayName = "VMnet5"
ethernet1.virtualDev = "e1000"
ethernet2.present = "TRUE"
ethernet1.present = "TRUE"
ethernet2.address = "00:50:56:25:B2:00"
ethernet1.address = "00:50:56:24:1A:79"
```

```
pfSense_UTM.vmx
파일 편집 보기
scsi0:0.fileName = "pfSense_UTM.vmdk"
scsi0:0.present = "TRUE"
ide1:0.deviceType = "cdrom-image"
ide1:0.fileName = "C:\VMware\SecurityNetwork\설치자료\WpfSense-CE-2.7.2-RELEASE-amd64.iso"
ide1:0.present = "TRUE"
usb.present = "TRUE"
ehci.present = "TRUE"
ethernet0.connectionType = "custom"
ethernet0.addressType = "static"
ethernet0.virtualDev = "vmxnet3"
ethernet0.present = "TRUE"
extendedConfigFile = "pfSense_UTM.vmx"
floppy0.present = "FALSE"
ethernet0.vnet = "VMnet8"
ethernet0.displayName = "VMnet8"
ethernet2.connectionType = "custom"
ethernet2.addressType = "static"
ethernet2.vnet = "VMnet6"
ethernet2.displayName = "VMnet6"
ethernet2.virtualDev = "vmxnet3"
ethernet1.connectionType = "custom"
ethernet1.addressType = "static"
ethernet1.vnet = "VMnet5"
ethernet1.displayName = "VMnet5"
ethernet1.virtualDev = "vmxnet3"
ethernet2.present = "TRUE"
ethernet1.present = "TRUE"
ethernet2.address = "00:50:56:25:B2:00"
ethernet1.address = "00:50:56:24:1A:79"
줄 45, 열 33 32/3,059자 일반 텍스트 100% Windows (CRLF) UTF-8
```

\\SecurityNetwork\Firewall\pfSense\pfSense_UTM.vmx

5) 네트워크 인터페이스 기능 지정

```
0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 1

Valid interfaces are:

vmx0      00:50:56:3a:9f:de (down) VMware VMXNET3 Ethernet Adapter
vmx1      00:50:56:24:1a:79 (down) VMware VMXNET3 Ethernet Adapter
vmx2      00:50:56:25:b2:00 (down) VMware VMXNET3 Ethernet Adapter

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y!n]? █
```

n → VLAN 기능 비활성화

네트워크 인터페이스 기능 지정

```
Should VLANs be set up now [y|n]? n
If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(vmx0 vmx1 vmx2 or a): vmx0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(vmx1 vmx2 a or nothing if finished): vmx1

Optional interface 1 description found: OPT1
Enter the Optional 1 interface name or 'a' for auto-detection
(vmx2 a or nothing if finished): vmx2

The interfaces will be assigned as follows:

WAN -> vmx0
LAN -> vmx1
OPT1 -> vmx2

Do you want to proceed [y|n]? y
```

- WAN 인터페이스 지정 : VMX0
- LAN 인터페이스 지정 : VM1
- Optional 인터페이스 지정 : OPT1

6) WAN 네트워크 인터페이스 설정과 IP 주소 지정

```
1 - WAN (vmx0 - dhcp)
2 - LAN (vmx1 - static)
3 - OPT1 (vmx2)

Enter the number of the interface you wish to configure: 1

Configure IPv4 address WAN interface via DHCP? (y/n) n

Enter the new WAN IPv4 address. Press <ENTER> for none:
> 192.168.10.254

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new WAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new WAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
> 192.168.10.2

Configure IPv6 address WAN interface via DHCP6? (y/n)
```

- 설정할 인터페이스 번호 지정 : 1
- WAN IP address : 192.168.10.254
- 서브넷마스크 : 24
- 게이트웨이 주소 : 192.168.10.2
- 그 외 IPv4/IPv6 DHCP 기능 비활성화

WAN 네트워크 인터페이스 설정과 IP 주소 지정

```
For a WAN, enter the new WAN IPv4 upstream gateway address.  
For a LAN, press <ENTER> for none:  
>
```

```
Configure IPv6 address WAN interface via DHCP6? (y/n) n
```

```
Enter the new WAN IPv6 address. Press <ENTER> for none:  
>
```

```
Do you want to enable the DHCP server on WAN? (y/n) n  
Disabling IPv4 DHCPD...  
Disabling IPv6 DHCPD...
```

```
Please wait while the changes are saved to WAN...  
Reloading filter...  
Reloading routing configuration...  
DHCPD...
```

```
The IPv4 WAN address has been set to 192.168.10.254/24  
You can now access the webConfigurator by opening the following URL in your web  
browser:  
http://192.168.10.254/
```

```
Press <ENTER> to continue. █
```


LAN/DMZ 네트워크 인터페이스 설정과 IP 주소 지정

```
Enter the number of the interface you wish to configure: 3
Configure IPv4 address OPT1 interface via DHCP? (y/n) n
Enter the new OPT1 IPv4 address. Press <ENTER> for none:
> 10.10.10.254
Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8
Enter the new OPT1 IPv4 subnet bit count (1 to 32):
> 24
For a WAN, enter the new OPT1 IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
Configure IPv6 address OPT1 interface via DHCP6? (y/n) n
Enter the new OPT1 IPv6 address. Press <ENTER> for none:
>
Do you want to enable the DHCP server on OPT1? (y/n) n
```

- 설정할 인터페이스 번호 지정 : 3
- WAN IP address : 10.10.10.254
- 서브넷마스크 : 24
- 그 외 IPv4/IPv6 DHCP 기능 비활성화

LAN/DMZ 네트워크 인터페이스 설정과 IP 주소 지정

```
For a WAN, enter the new OPT1 IPv4 upstream gateway address.  
For a LAN, press <ENTER> for none:  
>  
  
Configure IPv6 address OPT1 interface via DHCP6? (y/n) n  
  
Enter the new OPT1 IPv6 address. Press <ENTER> for none:  
>  
  
Do you want to enable the DHCP server on OPT1? (y/n) n  
Disabling IPv4 DHCPD...  
Disabling IPv6 DHCPD...  
  
Please wait while the changes are saved to OPT1...  
Reloading filter...  
Reloading routing configuration...  
DHCPD...  
  
The IPv4 OPT1 address has been set to 10.10.10.254/24  
You can now access the webConfigurator by opening the following URL in your web  
browser:  
http://10.10.10.254/  
  
Press <ENTER> to continue.
```

7) 네트워크 인터페이스 IP 주소 확인

```
The IPv4 OPT1 address has been set to 10.10.10.254/24
You can now access the webConfigurator by opening the following URL in your web
browser:
    http://10.10.10.254/

Press <ENTER> to continue.
VMware Virtual Machine - Netgate Device ID: abfef3ef2b25954e3af8

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> vmx0      -> v4: 192.168.10.254/24
LAN (lan)      -> vmx1      -> v4: 172.16.10.254/24
OPT1 (opt1)    -> vmx2      -> v4: 10.10.10.254/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: █
```

8) pfSense disable(비활성화)

```
VMware Virtual Machine - Netgate Device ID: abfef3ef2b25954e3af8

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

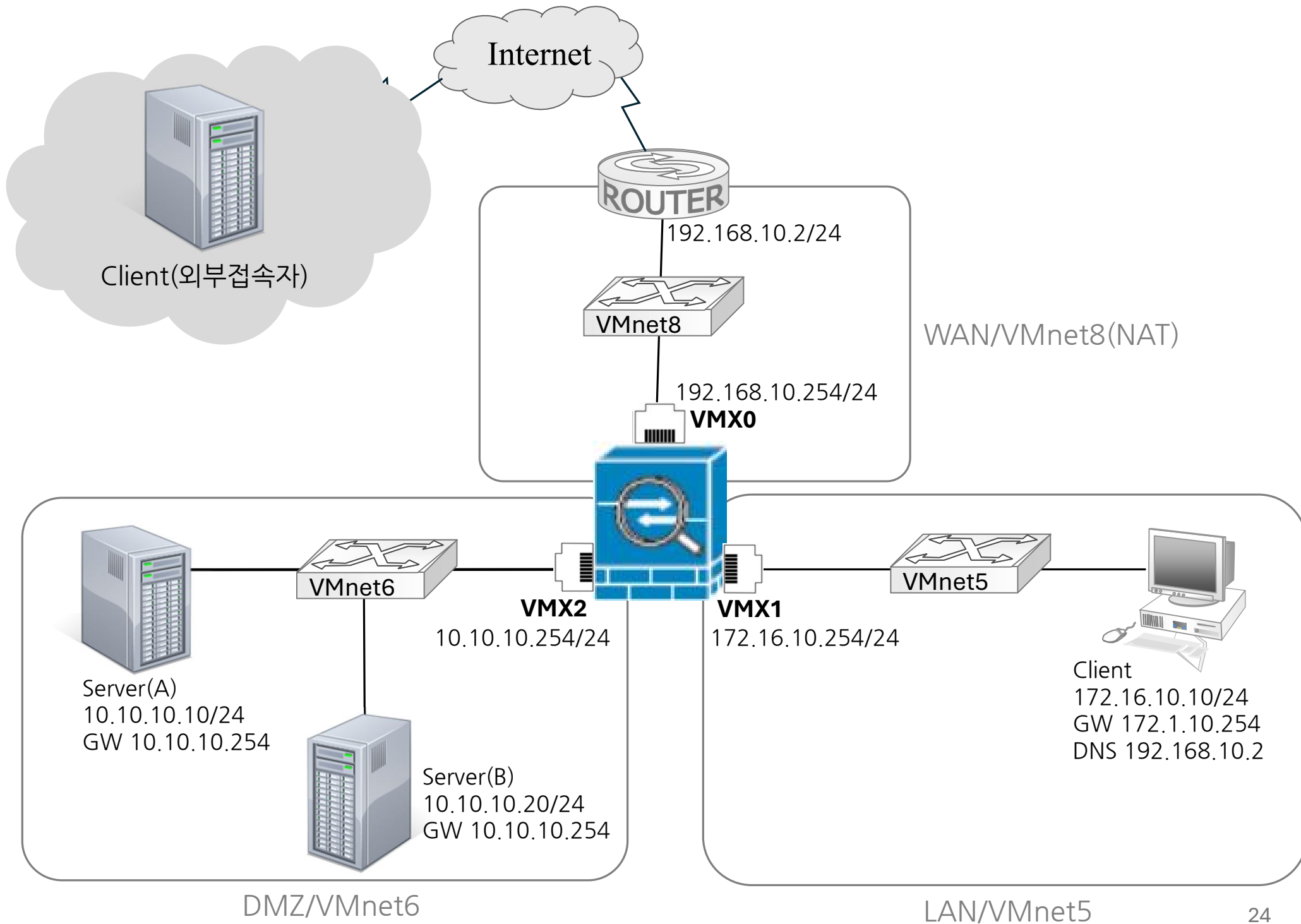
WAN (wan)      -> vmx0      -> v4: 192.168.10.254/24
LAN (lan)      -> vmx1      -> v4: 172.16.10.254/24
OPT1 (opt1)    -> vmx2      -> v4: 10.10.10.254/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

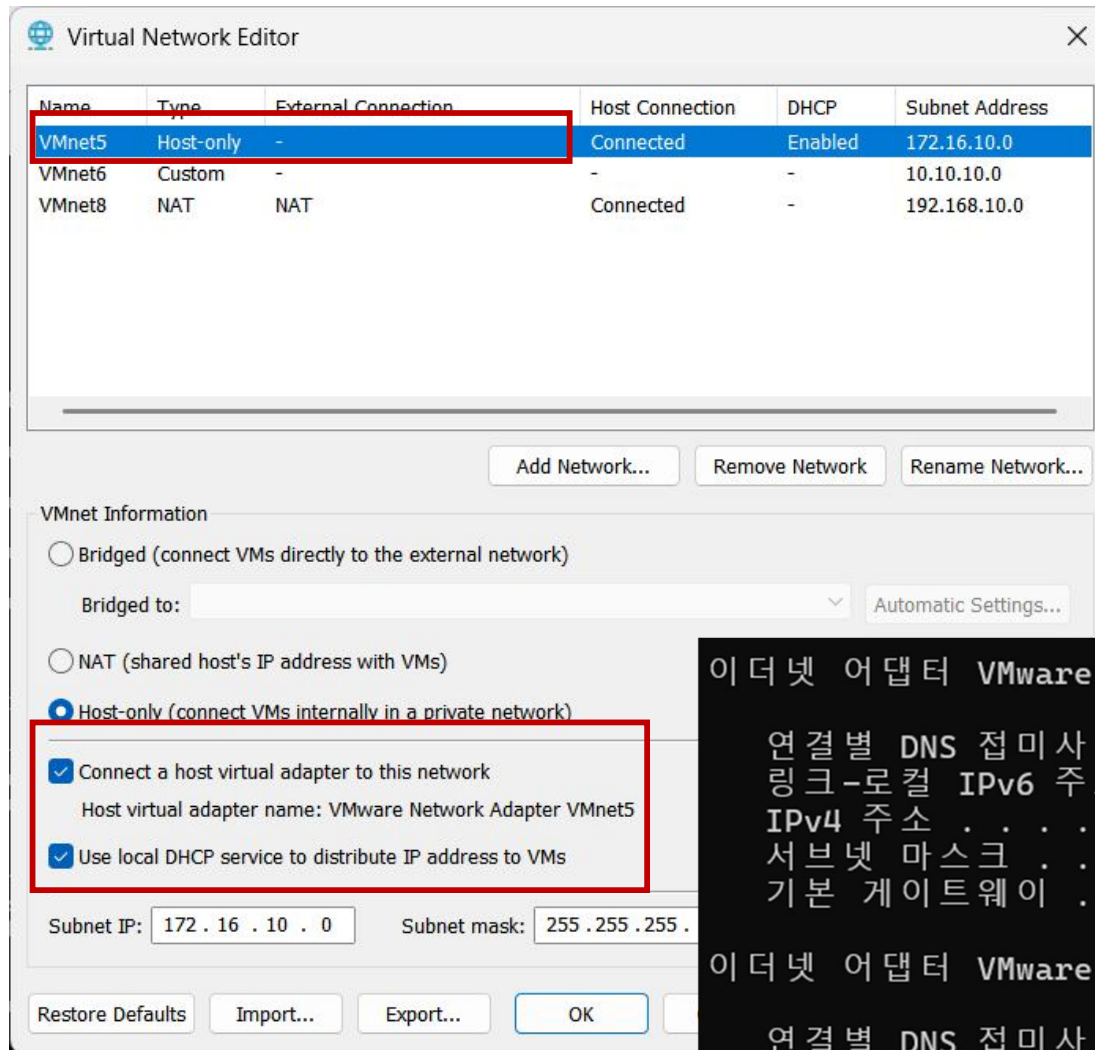
Enter an option: 8

[2.7.2-RELEASE][root@pfSense.home.arpal/root]: pfctl -d
pf disabled
```

pfctl -d



9) 방화벽 설정을 위한 네트워크 환경 변경



이더넷 어댑터 VMware Network Adapter VMnet8:

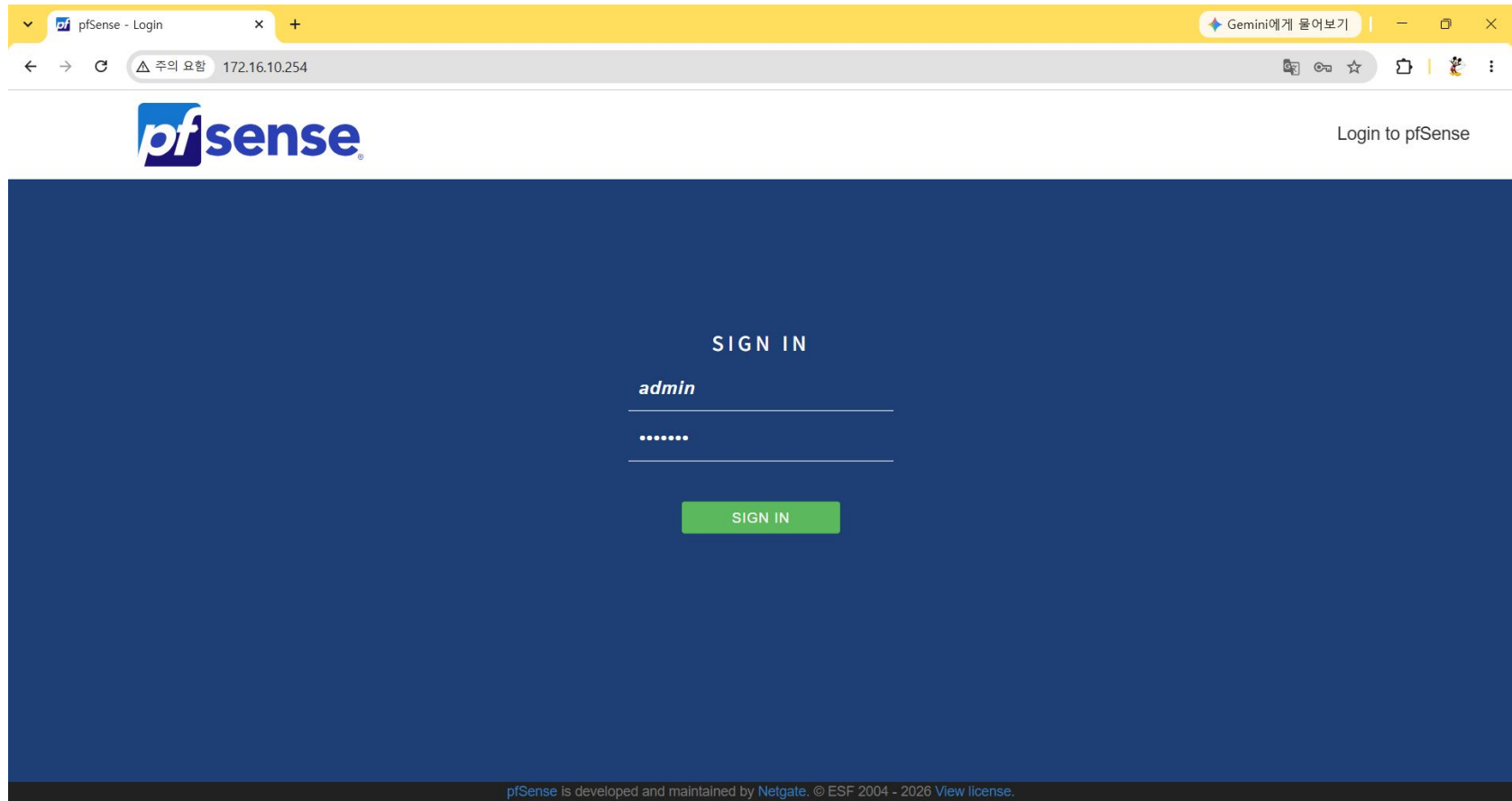
```
연결별 DNS 접미사 . . . . . :  
링크-로컬 IPv6 주소 . . . . . : fe80::52ff:3b7:f5d7:484%15  
IPv4 주소 . . . . . : 192.168.10.1  
서브넷 마스크 . . . . . : 255.255.255.0  
기본 게이트웨이 . . . . . :
```

이더넷 어댑터 VMware Network Adapter VMnet5:

```
연결별 DNS 접미사 . . . . . :  
링크-로컬 IPv6 주소 . . . . . : fe80::2143:cb26:2874:83f%55  
IPv4 주소 . . . . . : 172.16.10.1  
서브넷 마스크 . . . . . : 255.255.255.0  
기본 게이트웨이 . . . . . :
```

10) Web pfSense 접속

http://172.16.10.254



admin/pfsense

pfSens 명령어

- pfctl -d : 방화벽 rule 비활성화
- pfctl -e : 방화벽 rule 활성화
- pfctl -f /tmp/rules.debug : 방화벽 rule 강제 재로드
- pfctl -s state : 현재 활성화된 방화벽 상태 테이블 확인
- pfctl -F state : 상태 테이블 초기화
- ifconfig : 네트워크 인터페이스 정보 및 할당된 IP 주소 확인
- netstat -rn : 라우팅 테이블 확인
- exit : shell 상태에서 메뉴로 전환

Default 접근제어 Rule

Default LAN Rules

Firewall / Rules / LAN

Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	4/229 KiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	
☐	5/38 KiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	

Add Add Delete Toggle Copy Save Separator

현재세션수/누적트래픽양

*세션 수 : 실시간으로 접속을 요청한 세션수

*누적트래픽 양 : 해당 규칙이 적용되어 통과한 누적 트래픽 데이터 총량
(KiB, MiB, GiB 단위, 통신을 많이 할수록 증가)

Anti-Lockout Rule

- 접속 차단 방지 정책
- 관리자가 LAN 상에서 pfSense로 Web 및 SSH 접속을 무조건 허용하는 규칙
- 모든 트래픽 다찬 규칙을 넣더라도 WebUI접속은 항상 유지하게 함
- 보안상 비활성화가 필요한 경우
- System > Advanced > Admin Access > Anti-Lockout 항목 체크
- (Disable the webConfigurator anti-lockout rule)

* 해당 룰을 비활성화기 전 에 Web또는 ssh 접속을 반드시 허용하는 수동 방화벽 규칙을 만들어 뒀야 함

Anti-Lockout Rule 비활성화 방법

System / [Advanced](#) / [Admin Access](#)

[Admin Access](#) [Firewall & NAT](#) [Networking](#) [Miscellaneous](#) [System Tunables](#) [Notifications](#)

webConfigurator

Protocol

☒ HTTP ☐ HTTPS (SSL/TLS)

TCP port

Enter a custom port number for the webConfigurator above to override the default (80 for HTTP, 443 for HTTPS). Changes will take effect immediately after save.

WebGUI Login Autocomplete	☒ Enable webConfigurator login autocomplete When this is checked, login credentials for the webConfigurator may be saved by the browser. While convenient, some security standards require this to be disabled. Check this box to enable autocomplete on the login form so that browsers will prompt to save credentials (NOTE: Some browsers do not respect this option).
GUI login messages	☐ Lower syslog level for successful GUI login events When this is checked, successful logins to the GUI will be logged as a lower non-emergency level. Note: The console bell behavior can be controlled independently on the Notifications tab.
Roaming	☒ Allow GUI administrator client IP address to change during a login session When this is checked, the login session to the webConfigurator remains valid if the client source IP address changes.
Anti-lockout	☐ Disable webConfigurator anti-lockout rule When this is unchecked, access to the webConfigurator on the LAN interface is always permitted, regardless of the user-defined firewall rule set. Check this box to disable this automatically added rule, so access to the webConfigurator is controlled by the user-defined firewall rules (ensure a firewall rule is in place that allows access, to avoid being locked out!) <i>Hint: the "Set interface(s) IP address" option in the console menu resets this setting as well.</i>

WebUI 접속 허용 수동 방화벽 설정

- Action : Pass
- Interface : LAN
- Address Family : IPv4
- Protocol : TCP
- Source : Single host or alias (192.168.1.2)
- Destination : LAN address
- Destination Port Range : From HTTP / To HTTP
- 기본의 모든 허용 규칙 보다 무조건 위쪽에 위치해야 함

Default WAN Rules

Interfaces / WAN (vmx0)   

Static IPv4 Configuration

IPv4 Address

192.168.10.254

/ 24

IPv4 Upstream gateway

None

+ Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses

☒

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks

☒

Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

 Save

Firewall / Rules / WAN   

Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/692 B	*	RFC 1918 networks	*	*	*	*	*		Block private networks	
☒	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*		Block bogon networks	

Block private networks and loopback addresses

- RFC 1918(사설IP)와 루프백주소(127.0.0.0/8)
- 외부망(인터넷)에서 사설IP나 loopback IP를 송신지IP로 들어오는 것은
- 보안상 위험한 트래픽으로 간주하여 원천 차단한다는 규칙
- 해당 rule을 활성화 시 IP Spoofing이나 잘못된 라우팅 정책으로 내부로 들어오는 트래픽들을 차단 가능

Block bogon networks

- 인터넷에서 비정상 IP주소(bogon, 보곤) 접근을 차단
- Bogon 주소는 인터넷망에서 절대 사용될 수 없는 주소
 - 미할당 주소 : IANA에서 통신사나 기업에 아직 분배하지 않은 공인 IP 대역
 - 예약된 주소 : 테스트용, 연구용, 또는 특수목적(사설IP)으로 할당된 주소
 - 삭제된 주소 : 과거에 사용되었으나 현재는 회수되어 비어 있는 IP 대역
- 해당 rule 활성화 시 다양한 네트워크 공격을 차단할 수 있음(IP spoofing, DDoS 공격 등)
- Block private networks and loopback addresses 활성화는 단지 사설IP와 루프백 주소에만 초점을 맞춘 반면 Block bogon networks은 존재하지 않은 유령IP까지 포함하는 넓은 개념의 필터링

Block private networks /bogon network 비활성화

Interfaces / WAN (vmx0)

Static IPv4 Configuration

IPv4 Address: 192.168.10.254 / 24

IPv4 Upstream gateway: None [+ Add a new gateway](#)

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses ☐
Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks ☐
Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.



Firewall / Rules / WAN

Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
--------------------------	--------	----------	--------	------	-------------	------	---------	-------	----------	-------------	---------

No rules are currently defined for this interface
All incoming connections on this interface will be blocked until pass rules are added. Click the button to add a new rule.

[↑ Add](#) [↓ Add](#) [Delete](#) [Toggle](#) [Copy](#) [Save](#) [+ Separator](#)

State Table(상태 테이블 또는 세션 테이블)

연결된 세션 관계를 테이블에 기록해 놓고 접근 룰이 없어서 해당 세션관계의 트래픽들을 통과 시킴

Diagnostics / States / States ?

States Reset States

State Filter

Interface all

Filter expression Simple filter such as 192.168, v6, icmp or ESTABLISHED

Rule ID Comma separated list of integer rule IDs

Filter

States

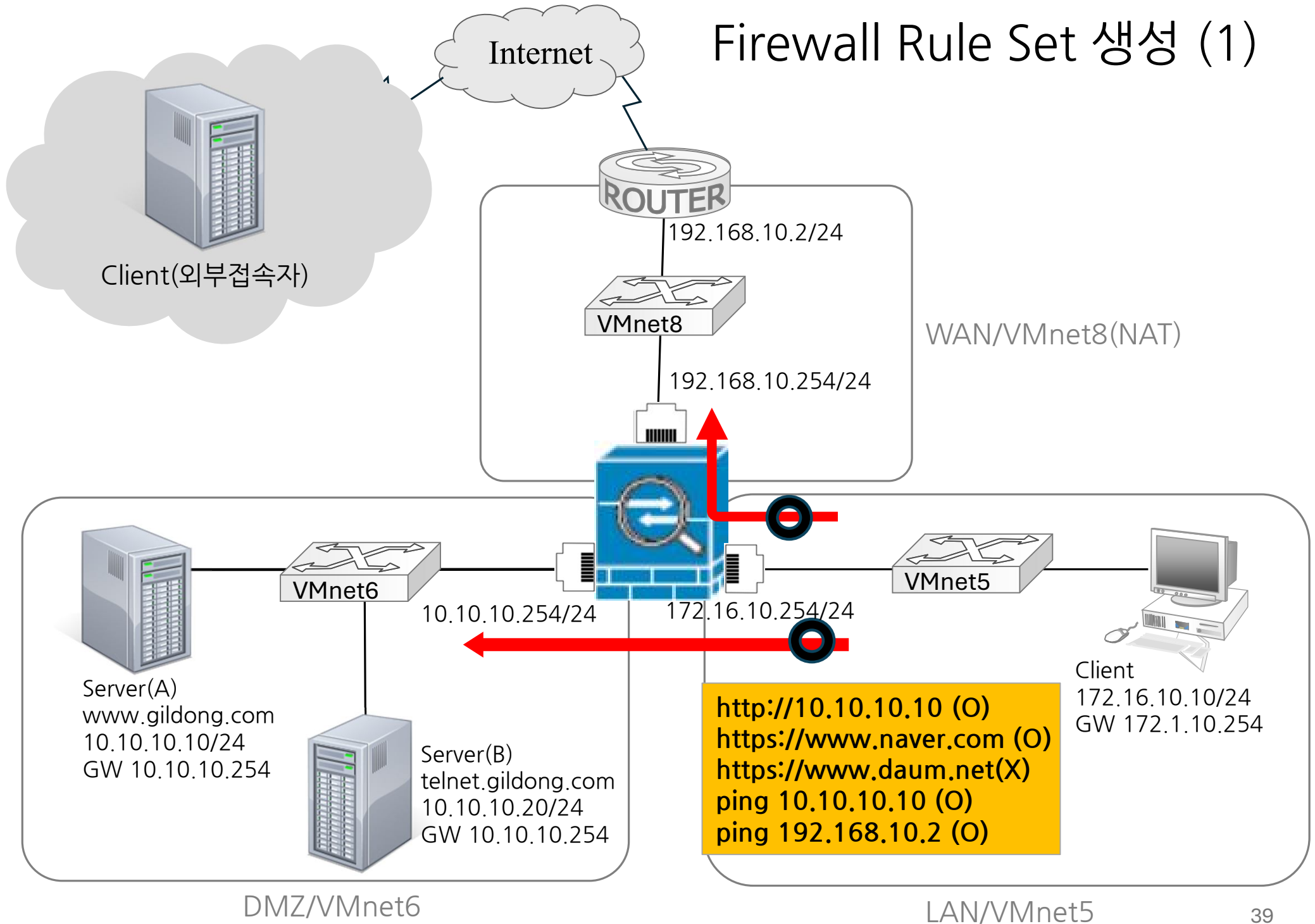
Interface	Protocol	Source (Original Source) -> Destination (Original Destination)	State	Packets	Bytes	
WAN	icmp	192.168.10.254:7248 -> 192.168.10.2:7248	0:0	17.979K / 17.979K	509 KiB / 509 KiB	
LAN	tcp	172.16.10.1:54967 -> 172.16.10.254:80	FIN_WAIT_2:ESTABLISHED	10 / 16	996 B / 12 KiB	
LAN	tcp	172.16.10.1:52417 -> 172.16.10.254:80	FIN_WAIT_2:ESTABLISHED	8 / 7	337 B / 352 B	
LAN	tcp	172.16.10.1:52885 -> 172.16.10.254:80	ESTABLISHED:ESTABLISHED	3 / 2	703 B / 92 B	
LAN	tcp	172.16.10.1:61509 -> 172.16.10.254:80	ESTABLISHED:ESTABLISHED	2 / 1	92 B / 52 B	

송신지 정보 → 수신지 정보

연결상태정보 전송량

Firewall Rule Set 생성

Firewall Rule Set 생성 (1)



Firewall Rule Set 생성 (1)

- 1) WAN/DMZ 망으로 ping 테스트 허용
- 2) DMZ 모든 서버들로 접속 허용
- 3) 외부망은 www.naver.com 외 모든 사이트 접속 차단

* Rule 생성 전 접근 가능 여부 테스트

https://www.naver.com (X)

https://www.daum.net(X)

http://10.10.10.10 (X)

http://10.10.10.20 (X)

ping 10.10.10.10 (X)

ping 192.168.10.2 (X)

1) WAN/DMZ 망으로 ping 테스트는 허용

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	ICMP
ICMP Subtypes	Any
Source	LAN subnets
Destination	Any
Log	Log packets that are handled by the rule

Edit Firewall Rule

Action Pass 

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface LAN 

Choose the interface from which packets must come to match this rule.

Address Family IPv4 

Select the Internet Protocol version this rule applies to.

Protocol ICMP 

Choose which IP protocol this rule should match.

ICMP Subtypes any 
Alternate Host
Datagram conversion error
Echo reply

For ICMP rules on IPv4, one or more of these ICMP subtypes may be specified.

Source

Source ☐ Invert match LAN subnets  Source Address / 

Destination

Destination ☐ Invert match Any  Destination Address / 

Extra Options

Log ☒ Log packets that are handled by this rule
Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).**Description**
A description may be entered here for administrative reference. A maximum of 52 characters will be used in the ruleset and displayed in the firewall log.**Advanced Options**  Display Advanced

2) DMZ 모든 서버들로 접속 허용

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	TCP
Source	LAN Subnets
Destination	DMZ Subnets
Destination Port Range	HTTP
Log	Log packets that are handled by the rule

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

TCP

Choose which IP protocol this rule should match.

Source

Source☐ Invert match

LAN subnets

Source Address

/

v



The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination☐ Invert match

DMZ subnets

Destination Address

/

v

Destination Port Range

HTTP (80)

From

Custom

HTTP (80)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log☒ Log packets that are handled by this rule

Hint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

3) 외부망은 www.naver.com 외 모든 사이트 접속 차단

- ① DNS 서버 접근 허용 룰 생성
- ② https://www.naver.com 접근 허용 룰 생성
- ③ 그 외 외부 사이트 접속 차단 룰 생성

① DNS 서버 접근 허용 룰 생성

1단계. 내부 DNS 서버 사용 설정

Service > DNS resolver > General Settings > General DNS Resolver Options

설정 필드	필드값
Enable	Enable DNS resolver
Network Interface	All
Outgoing Network Interfaces	All

ISC DHCP has reached end-of-life and will be removed in a future version of pfSense. Visit [System > Advanced > Networking](#) to switch DHCP backend.

[General Settings](#)[Advanced Settings](#)[Access Lists](#)

General DNS Resolver Options

Enable ☒ Enable DNS resolver

Listen Port

The port used for responding to DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 53.

Enable SSL/TLS Service

☐ Respond to incoming SSL/TLS queries from local clients

Configures the DNS Resolver to act as a DNS over SSL/TLS server which can answer queries from clients which also support DNS over TLS. Activating this option disables automatic interface response routing behavior, thus it works best with specific interface bindings.

SSL/TLS Certificate

The server certificate to use for SSL/TLS service. The CA chain will be determined automatically.

SSL/TLS Listen Port

The port used for responding to SSL/TLS DNS queries. It should normally be left blank unless another service needs to bind to TCP/UDP port 853.

Network Interfaces

WAN
LAN
DMZ
WANv6 Link Local

Interface IP addresses used by the DNS Resolver for responding to queries from clients. If an interface has both IPv4 and IPv6 addresses, both are

**Outgoing Network
Interfaces**

WAN
LAN
DMZ
WANv6 Link Local

Utilize different network interface(s) that the DNS Resolver will use to send queries to authoritative servers and receive their replies. By default all interfaces are used.

2단계. 외부 DNS 서버 사용 설정

System > General Setup > DNS Server Settings > DNS servers

System / General Setup

System

Hostname

pfSense

Name of the firewall host, without domain part.

Domain

home.arpa

Domain name for the firewall.

Do not end the domain name with '.local' as the final part (Top Level Domain, TLD). The 'local' TLD is widely used by mDNS (e.g. Avahi, Bonjour, Rendezvous, Airprint, Airplay) and some Windows systems and networked devices. These will not network correctly if the router uses 'local' as its TLD. Alternatives such as 'home.arpa', 'local.lan', or 'mylocal' are safe.

DNS Server Settings

DNS Servers

8.8.8.8

Address

Enter IP addresses to be used by the system for DNS resolution. These are also used for the DHCP service, DNS Forwarder and DNS Resolver when it has DNS Query Forwarding enabled.

DNS Hostname

Hostname

Enter the DNS Server Hostname for TLS Verification in the DNS Resolver (optional).

48

3단계. DNS 서버 접속 허용 룰 생성

Firewall > Rules > LAN

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	UDP
Source	LAN Subnets
Destination	Any
Destination Port Range	DNS(53)
Log	Log packets that are handled by the rule

Edit Firewall Rule

Action

Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

LAN

Choose the interface from which packets must come to match this rule.

Address Family

IPv4

Select the Internet Protocol version this rule applies to.

Protocol

UDP

Choose which IP protocol this rule should match.

Source

Source☐ Invert match

LAN subnets

Source Address

/

▼



The **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination☐ Invert match

Any

Destination Address

/

▼

Destination Port Range

DNS (53)

From

Custom

DNS (53)

To

Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

② https://www.naver.com 접근 허용 룰 생성

[Alias 지정] Firewall > Aliases

Firewall / Aliases / Edit ?

Properties

Name

NAVER_IP

The name of the alias may only consist of the characters "a-z, A-Z, 0-9 and _".

Description

A description may be entered here for administrative reference (not parsed).

Type

Host(s)

Host(s)

Hint

Enter as many hosts as desired. Hosts must be specified by their IP address or fully qualified domain name (FQDN). FQDN hostnames are periodically re-resolved and updated. If multiple IPs are returned by a DNS query, all are used. An IP range such as 192.168.1.1-192.168.1.10 or a small subnet such as 192.168.1.16/28 may also be entered and a list of individual IP addresses will be generated.

IP or FQDN

www.naver.com

Entry added Sun, 09 Aug 2026 15:53:42 +0000

설정 필드	필드값
Properties	Name : NAVER_IP Type : Host(s)
Hosts	IP or FQDN : www.naver.com

4단계. <https://www.naver.com> 접속 허용 룰 생성

Firewall > Rules > LAN

설정 필드	필드값
Action	Pass
Interface	LAN
Address Family	IPv4
Protocol	TCP
Source	LAN Subnets
Destination	Address or Alias : NAVER_IP
Destination Port Range	HTTPS(443)
Log	Log packets that are handled by the rule

Edit Firewall Rule

Action Pass

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled ☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface LAN

Choose the interface from which packets must come to match this rule.

Address Family IPv4

Select the Internet Protocol version this rule applies to.

Protocol TCP

Choose which IP protocol this rule should match.

Source

Source ☐ Invert match LAN subnets Source Address /  Display AdvancedThe **Source Port Range** for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, **any**.

Destination

Destination ☐ Invert match Address or Alias NAVER_IP / **Destination Port Range** HTTPS (443) HTTPS (443)
From Custom To Custom

Specify the destination port or port range for this rule. The "To" field may be left empty if only filtering a single port.

Extra Options

Log ☒ Log packets that are handled by this ruleHint: the firewall has limited local log space. Don't turn on logging for everything. If doing a lot of logging, consider using a remote syslog server (see the [Status: System Logs: Settings](#) page).

③ 그 외 외부 사이트 접속 차단 룰 생성

Firewall > Rules > LAN

설정 필드	필드값
Action	Block
Interface	LAN
Address Family	IPv4
Protocol	Any
Source	Any
Destination	Any
Log	Log packets that are handled by the rule

LAN Interface에 적용된 Rule Set

Firewall / Rules / LAN 📊 📋 ?

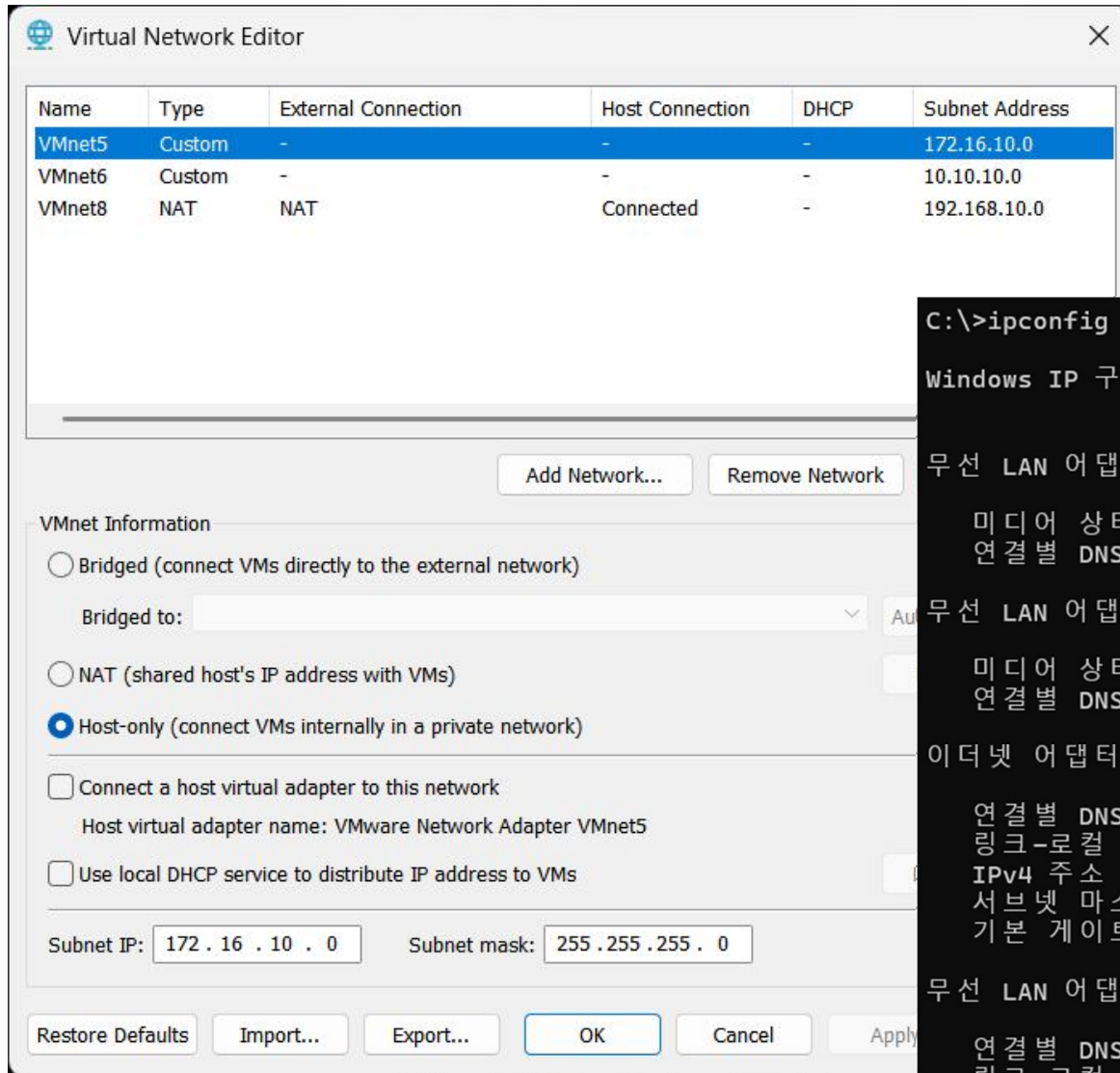
Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 1/551 KiB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☐	✓ 0/480 B	IPv4 ICMP any	LAN subnets	*	DMZ subnets	*	*	none			📌 ✎ 📄 🚫 🗑 ✕
☐	✓ 9/33 KiB	IPv4 UDP	LAN subnets	*	*	53 (DNS)	*	none			📌 ✎ 📄 🚫 🗑 ✕
☐	✓ 0/171 KiB	IPv4 TCP	LAN subnets	*	NAVER_IP	*	*	none			📌 ✎ 📄 🚫 🗑 ✕
☐	✓ 0/18 KiB	IPv4 TCP	LAN subnets	*	DMZ subnets	80 (HTTP)	*	none			📌 ✎ 📄 🚫 🗑 ✕
☐	✗ 0/66 KiB	IPv4 TCP	LAN subnets	*	*	*	*	none			📌 ✎ 📄 🚫 🗑

⬆ Add ⬇ Add 🗑 Delete 🚫 Toggle 📄 Copy 💾 Save ⬆ + Separator

테스트 환경 구성 (VMWare> Edit > Virtual Network Editor)



```
C:\>ipconfig

Windows IP 구성

무선 LAN 어댑터 로컬 영역 연결 * 1:

    미디어 상태 . . . . . : 미디어 연결 끊김
    연결별 DNS 접미사 . . . . . :

무선 LAN 어댑터 로컬 영역 연결 * 10:

    미디어 상태 . . . . . : 미디어 연결 끊김
    연결별 DNS 접미사 . . . . . :

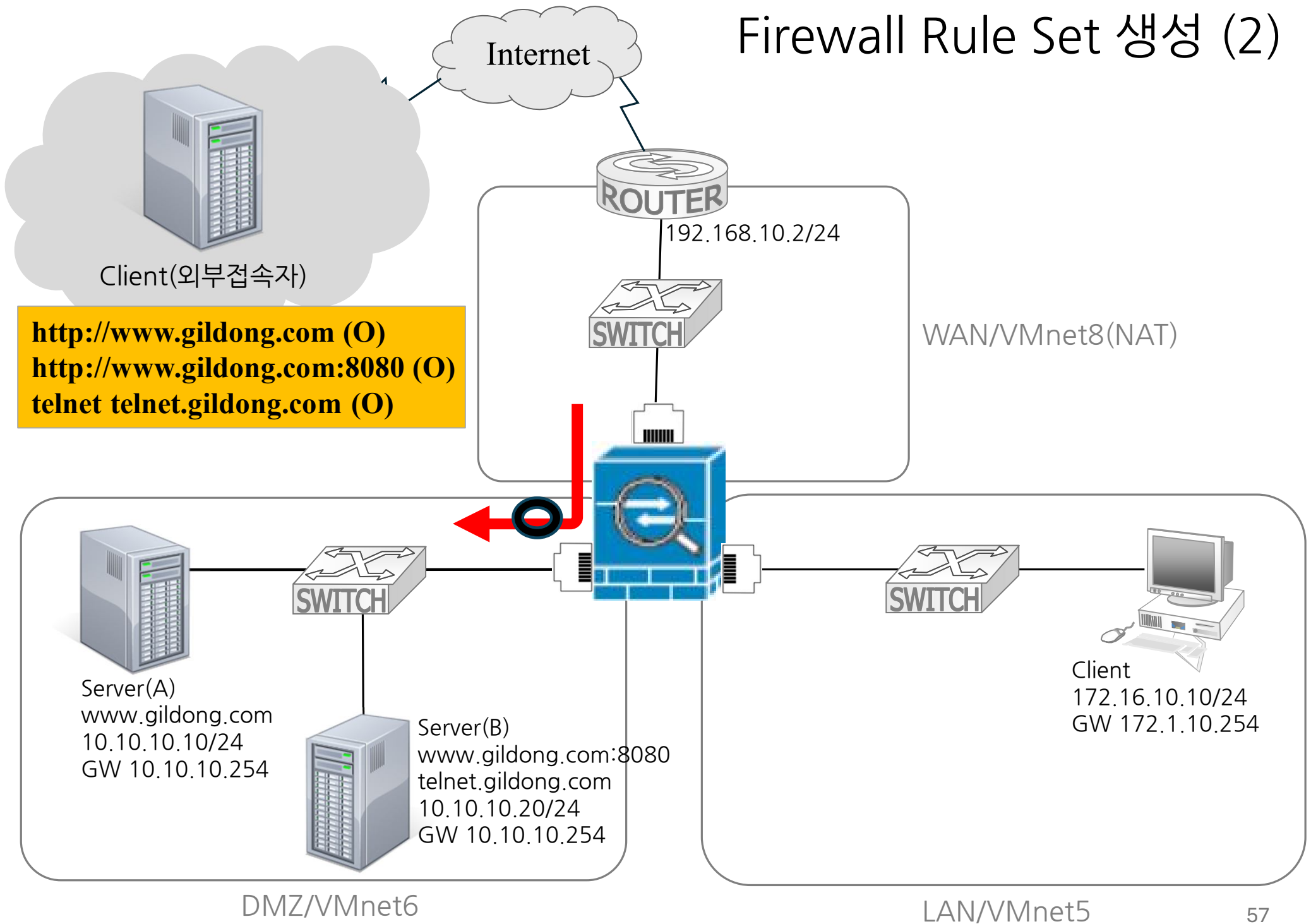
이더넷 어댑터 VMware Network Adapter VMnet8:

    연결별 DNS 접미사 . . . . . :
    링크-로컬 IPv6 주소 . . . . . : fe80::52ff:3b7:f5d7:484%16
    IPv4 주소 . . . . . : 192.168.10.1
    서브넷 마스크 . . . . . : 255.255.255.0
    기본 게이트웨이 . . . . . :

무선 LAN 어댑터 Wi-Fi:

    연결별 DNS 접미사 . . . . . :
    링크-로컬 IPv6 주소 . . . . . : fe80::6d01:1c27:487b:8dcb%19
    IPv4 주소 . . . . . : 172.18.10.198
    서브넷 마스크 . . . . . : 255.255.252.0
    기본 게이트웨이 . . . . . : 172.18.11.254
```

Firewall Rule Set 생성 (2)

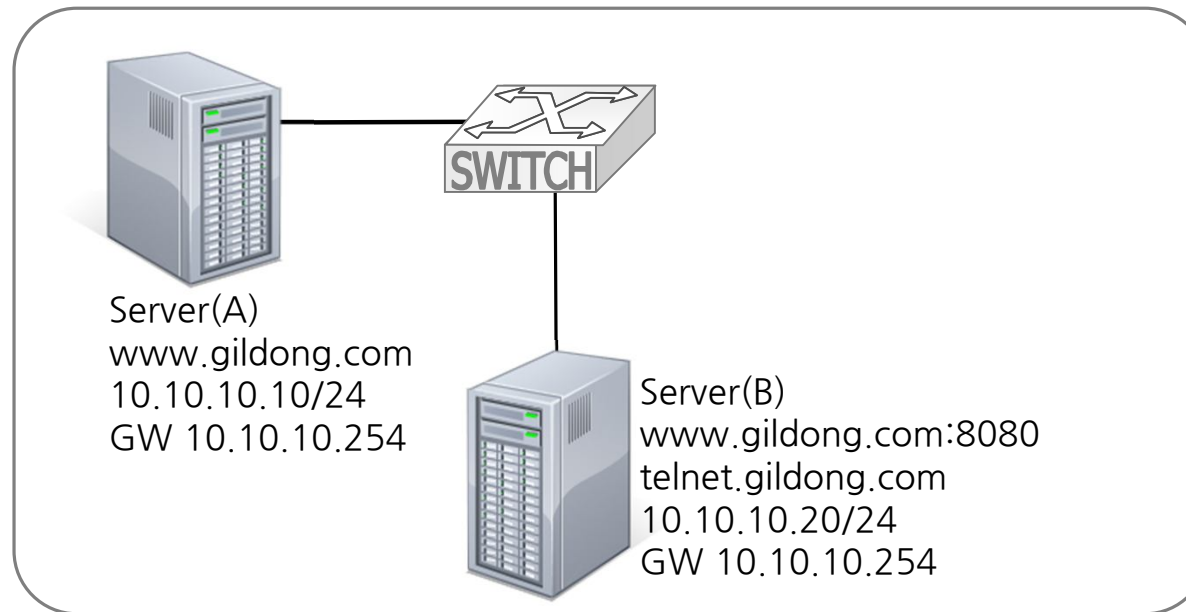


Firewall Rule Set 생성 (2)

- 1) 외부 접속자는 DMZ 서버로 접근 가능
- 2) 외부 접속자는 Telnet 서버로 접근 가능
- 3) 그 외 모든 접근은 차단

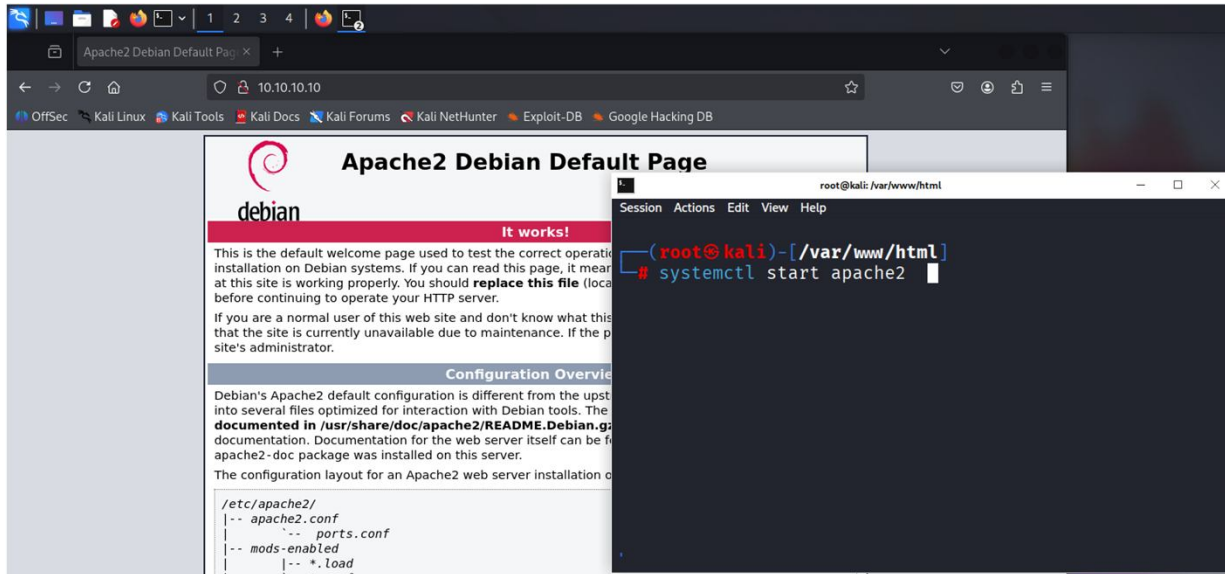
*** Rule 생성 전 접근 가능 여부 테스트**
http://www.gildong.com (X)
http://www.gildong.com:8080(X)
telnet telnet.gildong.com (X)

Firewall Rule Set 생성 (2)



외부주소	내부주소
192.168.10.254:80	10.10.10.10:80
192.168.10.254:8080	10.10.10.20:8080
192.168.10.254:23	10.10.10.20:23

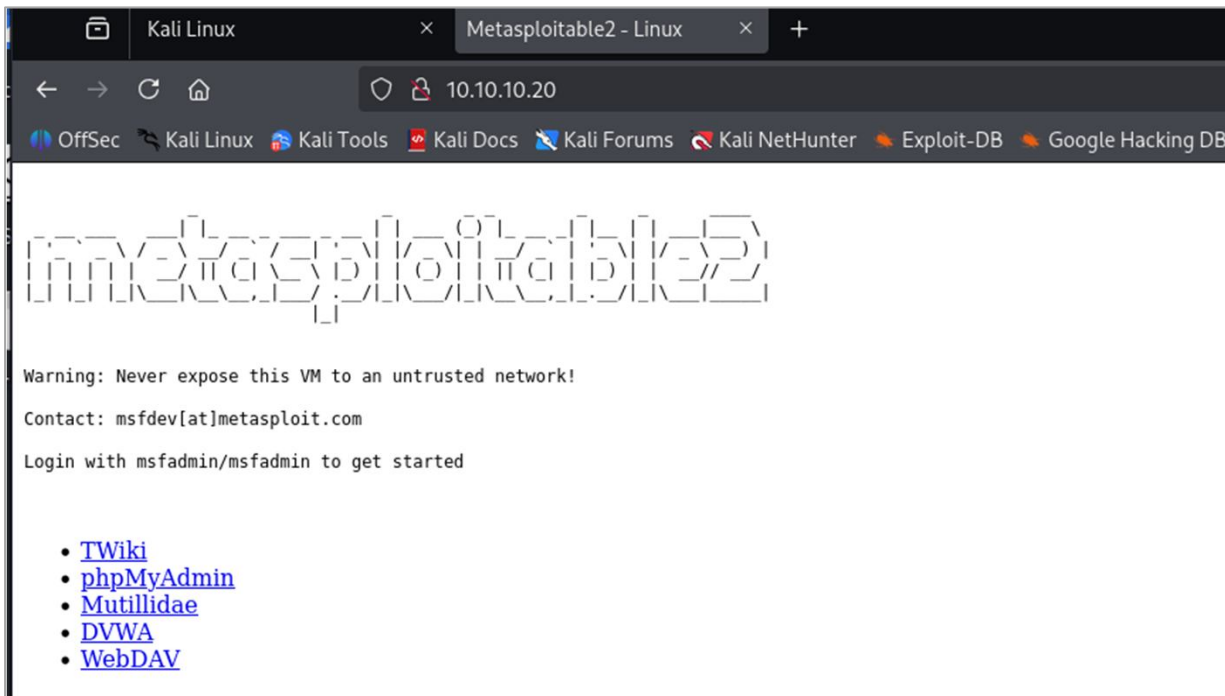
[DMZ Zone] Rule 생성 전 테스트 (웹서버 접속)



http://10.10.10.10

[WebServer 활성화]

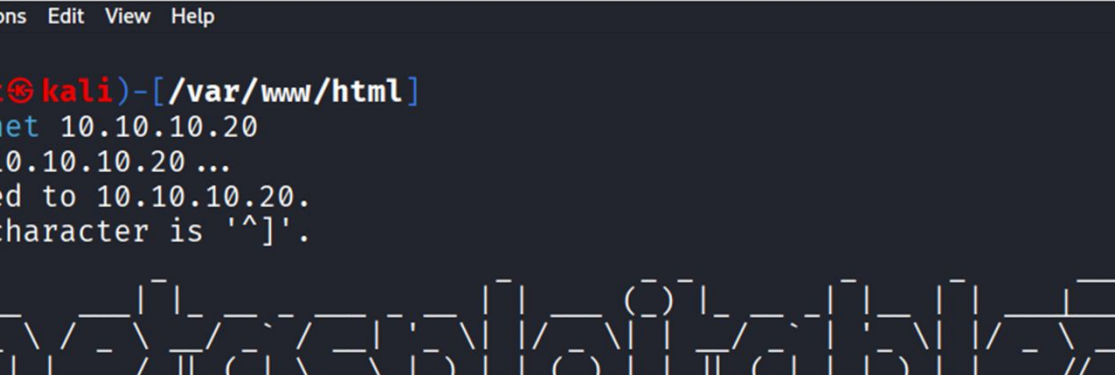
```
sudo su - root  
systemctl start apache2
```



http://10.10.10.20

[DMZ Zone] ule 생성 전 테스트 (텔넷 서버 접속)

```
telnet 10.10.10.20
```



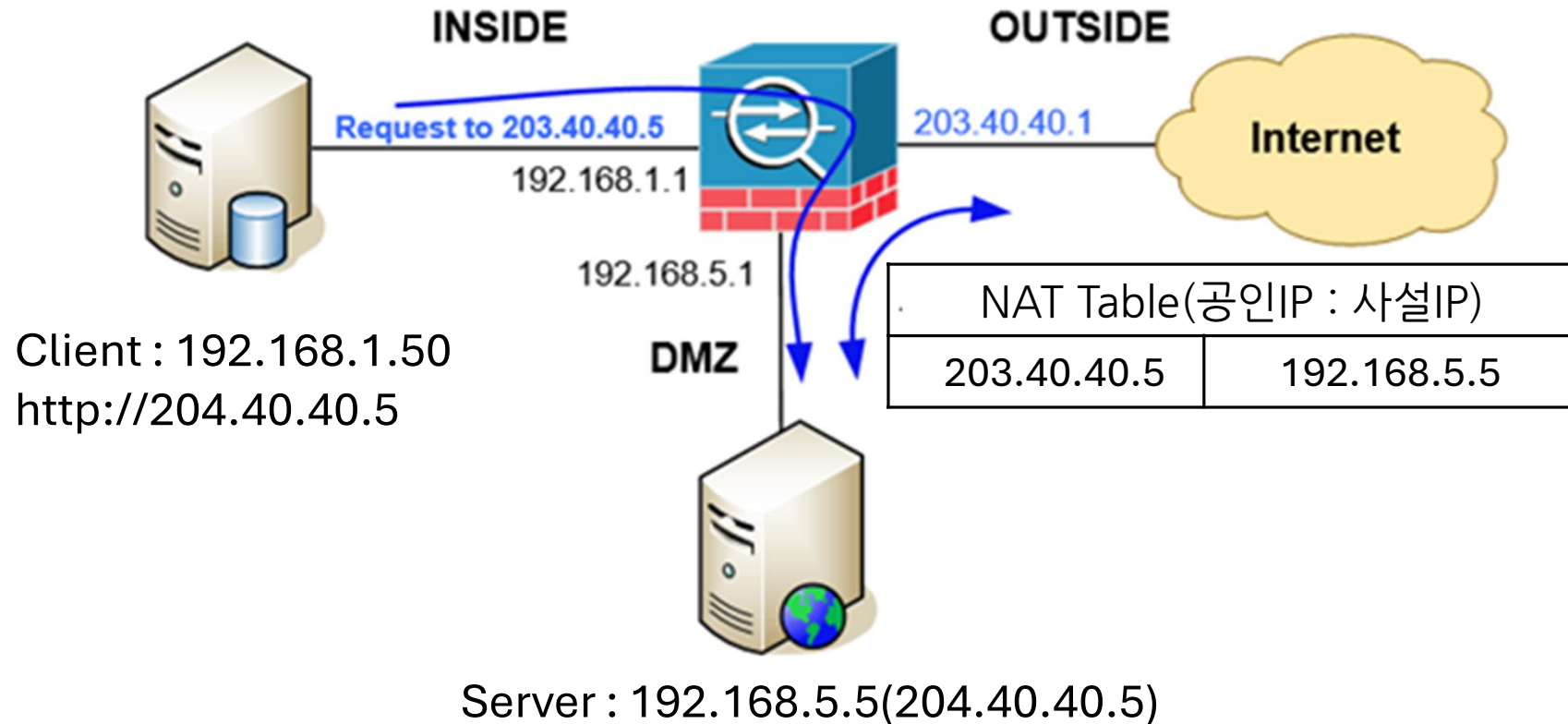
```
(root@kali) - [/var/www/html]
# telnet 10.10.10.20
Trying 10.10.10.20 ...
Connected to 10.10.10.20.
Escape character is '^]'.

Metasploit (msf5)

Warning: Never expose this VM to an untrusted network!
Contact: msfdev[at]metasploit.com
Login with msfadmin/msfadmin to get started

metasploitable login: 
```

1 단계. NAT Reflection 설정



<< NAT Reflection>>

- 내부망 서버이지만 해당 서버의 IP가 공인IP인 경우, 인터넷 상의 라우터로 패킷을 보내서 내부망 서버로 접속하는 것이 일반적인 형태
- 이러한 과정을 걸치지 않고 NAT table에 목적지 서버가 내부망 서버로 확인이 되면 인터넷 망 게이트웨이를 경유하지 않고 바로 내부망 서버로 접속을 가능케 함

System > Advanced > Firewall & NAT > Network Address Translation >

- NAT Reflection mode for port forward > Pure NAT로 변경
- Enable Automatic outbound NAT for Reflection 체크박스 활성화

System / Advanced / Firewall & NAT

Admin Access

Firewall & NAT

Networking

Miscellaneous

System Tunables

Notifications

Network Address Translation

NAT Reflection mode for port forwards

Pure NAT

The Pure NAT mode uses a set of NAT rules to direct packets to the target of the port forward. It has better scalability, but it must be possible to accurately determine the interface and gateway IP used for communication with the target at the time the rules are loaded. There are no inherent limits to the number of ports other than the limits of the protocols. All protocols available for port forwards are supported.

- The NAT + Proxy mode uses a helper program to send packets to the target of the port forward. It is useful in setups where the interface and/or gateway IP used for communication with the target cannot be accurately determined at the time the rules are loaded. Reflection rules are not created for ranges larger than 500 ports and will not be used for more than 1000 ports total between all port forwards. This feature does not support IPv6. Only TCP and UDP protocols are supported.

Individual rules may be configured to override this system setting on a per-rule basis.

Reflection Timeout

2000

Enter value for Reflection timeout in seconds.
Note: Only applies to Reflection on port forwards in NAT + proxy mode.

Enable NAT Reflection for 1:1 NAT

☐ Automatic creation of additional NAT redirect rules from within the internal networks.
Note: Reflection on 1:1 mappings is only for the inbound component of the 1:1 mappings. This functions the same as the pure NAT mode for port forwards. For more details, refer to the pure NAT mode description above. Individual rules may be configured to override this system setting on a per-rule basis.

Enable automatic outbound NAT for Reflection

☒ Automatic create outbound NAT rules that direct traffic back out to the same subnet it originated from.
Required for full functionality of the pure NAT mode of NAT Reflection for port forwards or NAT Reflection for 1:1 NAT. Note: This only works for assigned interfaces. Other interfaces require manually creating the outbound NAT rules that direct the reply packets back through the router.

TFTP Proxy

WAN
LAN
DMZ

Choose the interfaces on which to enable TFTP proxy helper.

2단계. Port Forwarding 설정(1)

[Web Server(A) NAT 구성] Firewall > NAT > Port Forwarding

설정 필드	필드값
Interface	WAN
Address Family	IPv4
Interface	WAN
Destination	WAN address
Destination Port Range	HTTP
Redirect Target IP	Address or Alias 10.10.10.10
Redirect Target Port	HTTP
Filter rule association	Add associated filter rule

Edit Redirect Entry

Disabled ☐ Disable this rule

No RDR (NOT) ☐ Disable redirection for traffic matching this rule

This option is rarely needed. Don't use this without thorough knowledge of the implications.

Interface

Choose which interface this rule applies to. In most cases "WAN" is specified.

Address Family

Select the Internet Protocol version this rule applies to.

Protocol

Choose which protocol this rule should match. In most cases "TCP" is specified.

Destination ☐ Invert match. /

Type

Address/mask

Destination port range

From port

Custom

To port

Custom

Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.

Redirect target IP

Type

Address

Enter the internal IP address of the server on which to map the ports. e.g.: 192.168.1.12 for IPv4
In case of IPv6 addresses, it must be from the same "scope",
i.e. it is not possible to redirect from link-local addresses scope (fe80:*) to local scope (::1)

Redirect target port

Port

Custom

Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).
This is usually identical to the "From port" above.

Description

A description may be entered here for administrative reference (not parsed).

No XMLRPC Sync ☐ Do not automatically sync to other CARP members

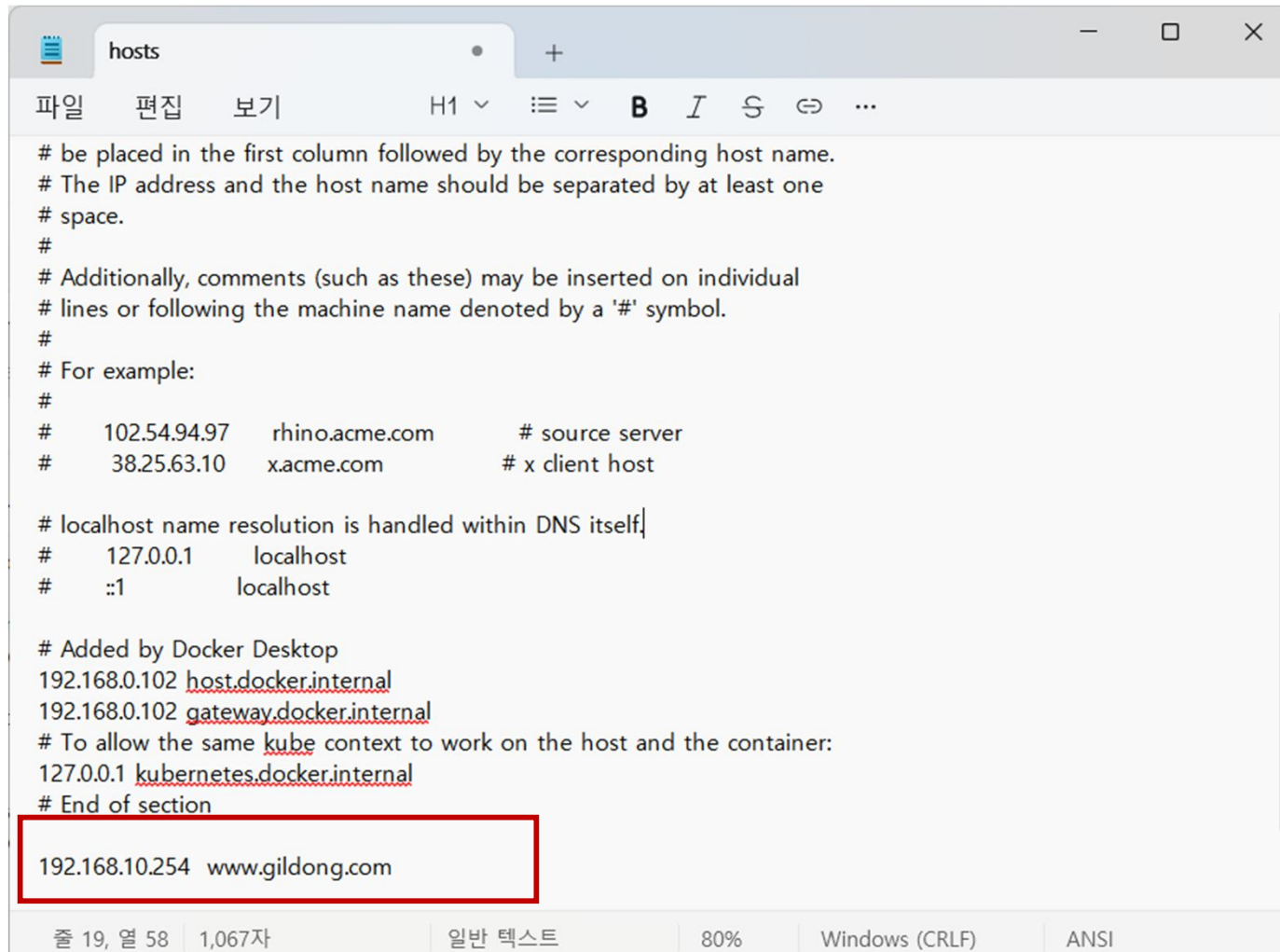
This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.

NAT reflection

Filter rule association

The "pass" selection does not work properly with Multi-WAN. It will only work on an interface containing the default gateway.

외부접속자 PC 환경 변경



```
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
# 102.54.94.97 rhino.acme.com # source server
# 38.25.63.10 x.acme.com # x client host

# localhost name resolution is handled within DNS itself
# 127.0.0.1 localhost
# ::1 localhost

# Added by Docker Desktop
192.168.0.102 host.docker.internal
192.168.0.102 gateway.docker.internal
# To allow the same kube context to work on the host and the container:
127.0.0.1 kubernetes.docker.internal
# End of section

192.168.10.254 www.gildong.com
```

C:\Windows\System32\drivers\etc\host
*(관리자 권한으로 실행)

Port Forwarding 설정 후 WAN Rule 자동 등록

Firewall / NAT / Port Forward

Port Forward 1:1 Outbound NPt

Rules

☐	Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☐	☒ WAN	TCP	*	*	WAN address	80 (HTTP)	10.10.10.10	80 (HTTP)		

Add Add Delete Toggle Save Separator



자동 등록됨

Firewall / Rules / WAN

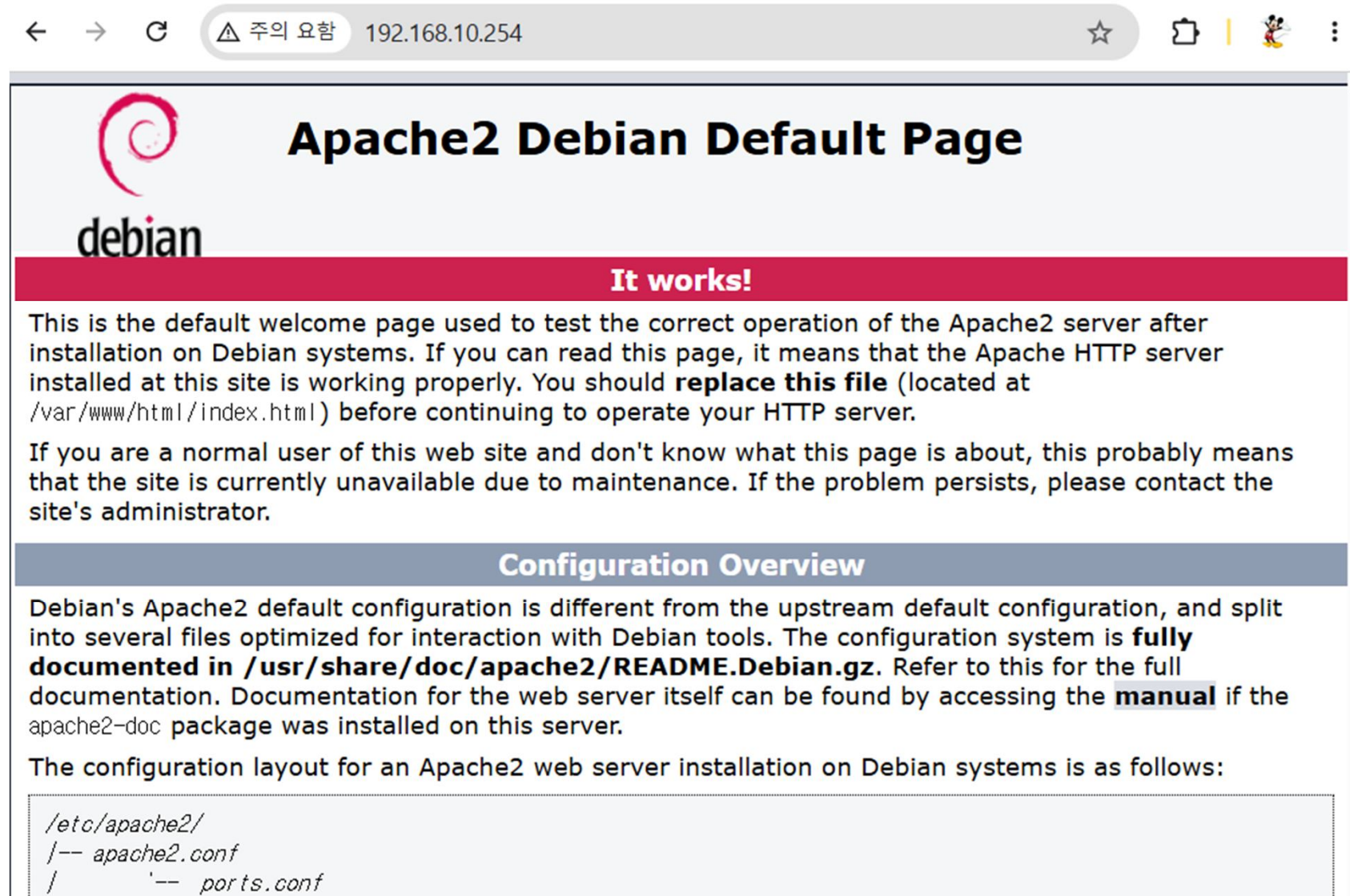
Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	☒ 0/28 KiB	IPv4 TCP	*	*	10.10.10.10	80 (HTTP)	*	none		NAT	

Add Add Delete Toggle Copy Save Separator

[크롬] http://www.gildong.com 또는 http://192.168.10.254



2단계. Port Forwarding 설정(2)

[Web Server(B) NAT 구성] Firewall > NAT > Port Forwarding

설정 필드	필드값
Interface	WAN
Address Family	IPv4
Protocol	TCP
Destination	WAN address
Destination Port Range	Other 8080
Redirect Target IP	Address or Alias 10.10.10.20
Redirect Target Port	HTTP
Filter rule association	Add associated filter rule

Edit Redirect Entry

Disabled ☐ Disable this rule

No RDR (NOT) ☐ Disable redirection for traffic matching this rule

This option is rarely needed. Don't use this without thorough knowledge of the implications.

Interface

Choose which interface this rule applies to. In most cases "WAN" is specified.

Address Family

Select the Internet Protocol version this rule applies to.

Protocol

Choose which protocol this rule should match. In most cases "TCP" is specified.

Source [Display Advanced](#)

Destination ☐ Invert match.

Type

Address/mask

Destination port range

From port

Custom

To port

Custom

Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.

Redirect target IP

Type

Address

Redirect target port

Port

Custom

Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).

This is usually identical to the "From port" above.

Description

A description may be entered here for administrative reference (not parsed).

No XMLRPC Sync ☐ Do not automatically sync to other CARP members

This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.

NAT reflection

Filter rule association

[View the filter rule](#)

Firewall / NAT / Port Forward



Port Forward

1:1

Outbound

NPt

Rules

☐		Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☐	☒	☒	WAN	TCP	*	*	WAN address	80 (HTTP)	10.10.10.10	80 (HTTP)	
☐	☒	☒	WAN	TCP	*	*	WAN address	8080	10.10.10.20	80 (HTTP)	

Add
 Add
 Delete
 Toggle
 Save
 Separator



자동 등록됨

Firewall / Rules / WAN



Floating

WAN

LAN

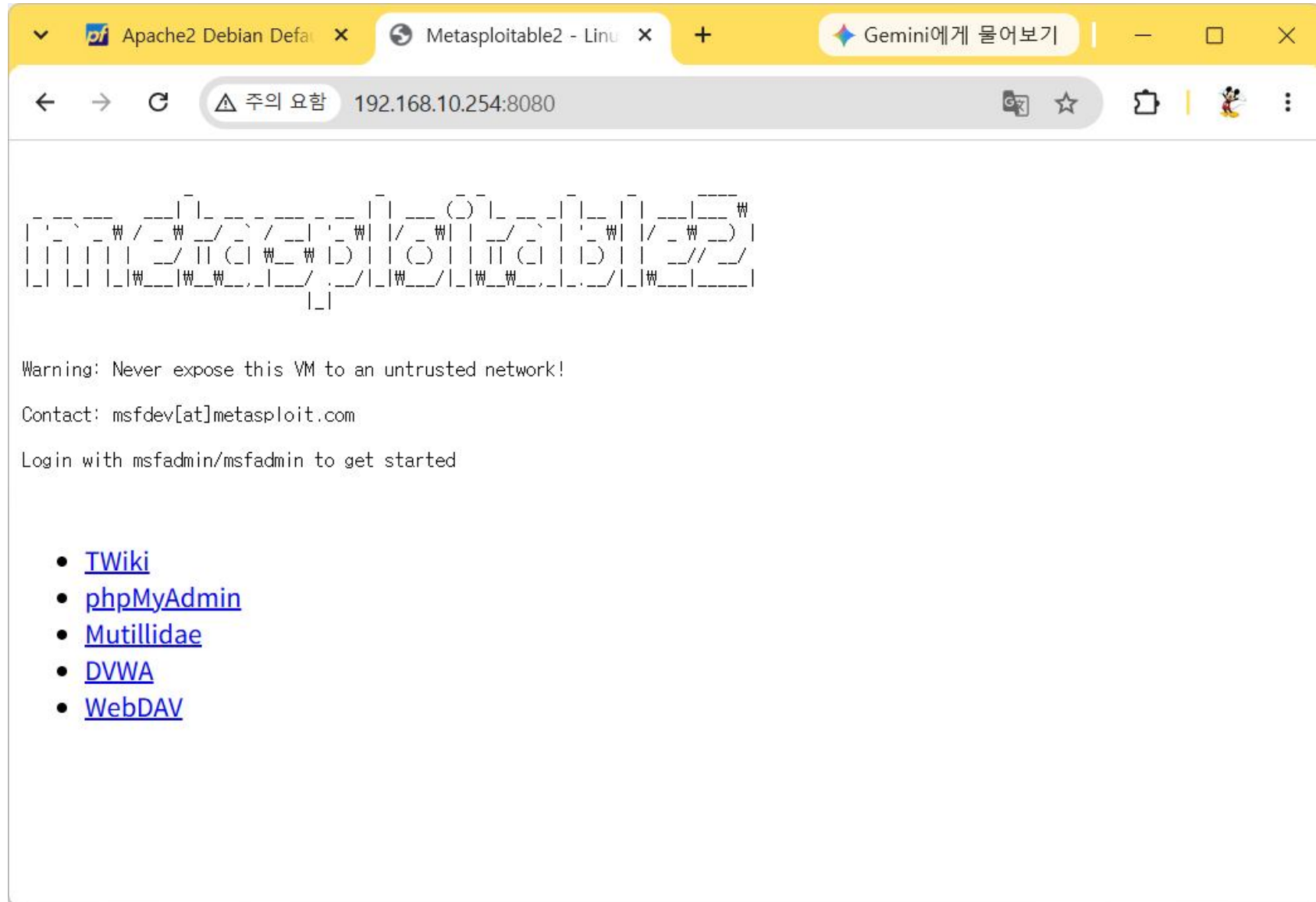
DMZ

Rules (Drag to Change Order)

☐		States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	☒	0/12 KiB	IPv4 TCP	*	*	10.10.10.10	80 (HTTP)	*	none		NAT	
☐	☒	0/4 KiB	IPv4 TCP	*	*	10.10.10.20	80 (HTTP)	*	none		NAT	

Add
 Add
 Delete
 Toggle
 Copy
 Save
 Separator

[크롬] <http://www.gildong.com:8080> 또는 <http://192.168.10.254:8080>



3단계. Port Forwarding 설정(3)

[Telnet Server NAT 구성] Firewall > NAT > Port Forwarding

설정 필드	필드값
Interface	WAN
Address Family	IPv4
Protocol	TCP
Destination	WAN address
Destination Port Range	Telnet
Redirect Target IP	Address or Alias 10.10.10.20
Redirect Target Port	Telnet
Filter rule association	Add associated filter rule

Edit Redirect Entry

Disabled ☐ Disable this rule

No RDR (NOT) ☐ Disable redirection for traffic matching this rule

This option is rarely needed. Don't use this without thorough knowledge of the implications.

Interface WAN

Choose which interface this rule applies to. In most cases "WAN" is specified.

Address Family IPv4

Select the Internet Protocol version this rule applies to.

Protocol TCP

Choose which protocol this rule should match. In most cases "TCP" is specified.

Source Display Advanced

Destination ☐ Invert match.

WAN address

Type

/

Address/mask

Destination port range Telnet

From port

Custom

Custom

Telnet

To port

Custom

Custom

Specify the port or port range for the destination of the packet for this mapping. The 'to' field may be left empty if only mapping a single port.

Redirect target IP Address or Alias

Type

10.10.10.20

Address

Redirect target port Telnet

Port

Custom

Specify the port on the machine with the IP address entered above. In case of a port range, specify the beginning port of the range (the end port will be calculated automatically).

This is usually identical to the "From port" above.

Description

A description may be entered here for administrative reference (not parsed).

No XMLRPC Sync ☐ Do not automatically sync to other CARP members

This prevents the rule on Master from automatically syncing to other CARP members. This does NOT prevent the rule from being overwritten on Slave.

NAT reflection Use system default

Filter rule association Add associated filter rule

The "pass" selection does not work properly with Multi-WAN. It will only work on an interface containing the default gateway.

4단계. 그 외 모든 접근은 차단

- Firewall > Rules > WAN

설정 필드	필드값
Action	Block
Interface	WAN
Address Family	IPv4
Protocol	Any
Source	Any
Destination	Any

Edit Firewall Rule

Action

Block ▼

Choose what to do with packets that match the criteria specified below.

Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.

Disabled☐ Disable this rule

Set this option to disable this rule without removing it from the list.

Interface

WAN ▼

Choose the interface from which packets must come to match this rule.

Address Family

IPv4 ▼

Select the Internet Protocol version this rule applies to.

Protocol

Any ▼

Choose which IP protocol this rule should match.

Source

Source☐ Invert match

Any ▼

Source Address

/ ▼

Destination

Destination☐ Invert match

Any ▼

Destination Address

/ ▼

WAN Interface에 적용된 Rule Set

Firewall / Rules / WAN

The changes have been applied successfully. The firewall rules are now reloading in the background.
[Monitor](#) the filter reload progress.

Floating **WAN** LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/12 KiB	IPv4 TCP	*	*	10.10.10.10	80 (HTTP)	*	none		NAT	    
☐	✓ 0/4 KiB	IPv4 TCP	*	*	10.10.10.20	80 (HTTP)	*	none		NAT	    
☐	✓ 0/0 B	IPv4 TCP	*	*	10.10.10.20	23 (Telnet)	*	none		NAT	    
☐	✗ 0/0 B	IPv4 *	*	*	*	*	*	none			    

 Add  Add  Delete  Toggle  Copy  Save  Separator

접근 테스트

